Reliability-Gated Visual-Sensor-State Fusion with Learned Fallback for Waste Classification

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Abstract– Waste classifiers can combine appearance with a structured auxiliary state, but that state may become missing, noisy, biased, or spatially stuck. We present a reliability-gated framework with a learned sensor-state fallback. Visual and class-text context predict a fallback latent and reference grid; record availability, raw-grid cues, cross-modal consistency, latent discrepancy, and bounded variance then control its archived residual. Evaluation uses an internal 19-class benchmark of real RGB images paired with normalized 4×4 sidecars. Current provenance records 8,639 image-conditioned vision-language estimates and 15 deterministic fallbacks, so independent hardware sensing is not claimed. Across three seeds, the method obtains $88.26\% \pm 0.21\%$ Accuracy and $82.26\% \pm 0.31\%$ Macro F1, improving on adapted QMF by 3.47 and 3.89 points; Visual-only remains the strongest clean classifier at 88.98/83.95%. Under predefined low light, the proposed model reaches 32.77/27.06%, whereas forcing the same-checkpoint sidecar missing or globally permuting it reduces Macro F1 to 24.21% and 26.21%. The compound reliability/fallback subsystem contributes 0.81 Accuracy and 0.51 Macro-F1 points under matched retraining. A separate constant-gate retraining control finds no consistent average classification advantage for sample-adaptive r, bounding its role to fault-sensitive quality ordering and selective robustness. For available corruptions, direct gate AUROC/AUPRC is $0.704 \pm 0.013/0.710 \pm 0.007$; absence is handled deterministically. A class-memory probe supports class-aligned retrieval, but nonsemantic class-ID keys match or exceed text-derived keys.

Index Terms– waste classification; multimodal fusion; structured sidecar; reliability gating; missing modality

I. INTRODUCTION

Automated waste recognition supports smart bins, sorting lines, and material-recovery systems. WasteNet demonstrated edge-oriented visual classification [1], while TACO, ZeroWaste, SortWaste, and ScatSpotter exposed natural context, clutter, dense scenes, and collection-distribution effects [2] [3] [4] [5]. These studies motivate task-aligned evaluation protocols and explicit robustness analysis when observation quality changes.

Visible appearance alone is often ambiguous for waste objects that differ in material composition, moisture, contamination, or internal structure. Independent physical sensing can provide complementary evidence, and prior systems combine RGB with hyperspectral, thermal, or terahertz channels [6]; SpectralWaste likewise provides synchronized RGB and hyperspectral facility data [7]. The archive evaluated here is different: each RGB object is associated with a normalized 4×4 sensor-state sidecar and five scalar sidecar attributes named weight, temperature, moisture, conductivity, and odor response. The classifier uses only the spatial map. Current record provenance identifies the sidecars as image-conditioned estimates rather than synchronized hardware measurements, so the experiments test reliability control at this structured interface rather than physical-modality independence.

This work studies object-level waste classification from real RGB images paired by 'sample_id' with normalized 4×4 sensor-state records. The 19-class internal benchmark contains 8,654 annotated objects from 5,990 source images. The training and evaluation interface preserves the 4×4 row-column structure and groups partitions by 'source_image_id'; Sections III-C and IV-B specify the exact record-to-tensor path and its provenance boundary.

Robust fusion remains necessary for a structured sidecar interface. A record can be unavailable, numerically corrupted, directionally biased, or reduced to a stuck spatial pattern. Direct concatenation cannot distinguish an internally consistent record from a failed one, and a classifier trained only on complete clean inputs can over-rely on the auxiliary branch. A useful model should estimate whether the current record is usable, attenuate it when evidence quality deteriorates, and preserve a defined prediction path when the record is absent.

Multimodal transformers are not inherently robust to missing inputs [8], and broader diagnostic studies likewise find that nominally strong multimodal models can become dominated by one channel [9]. Existing approaches include learning from severely incomplete modality sets [10], quality-aware fusion [11], representation recovery or distillation [12] [13], and aleatoric uncertainty modeling [14]. However, the present setting requires these ideas to operate jointly on a compact spatial state: the model must recover a missing representation, compare predicted and archived structure, and regulate the contribution of the archived residual within one end-to-end classifier.

We address this problem with a reliability-gated multimodal classifier. A YOLO11 visual backbone extracts multiscale object features [15], and CLIP category-text prototypes provide semantic context [16]. Their joint representation predicts both a fallback sensor-state latent and a reference spatial grid. In parallel, a sensor encoder maps the archived 4×4 sidecar into the same latent space. A sample-wise gate combines availability, raw-grid quality, predicted-observed grid consistency, latent discrepancy, and a bounded variance diagnostic. The effective sensor-state representation is the predicted latent plus a reliability-scaled archived residual. This construction moves continuously among retaining a consistent record, attenuating a degraded one, and reverting to the predicted state when the sidecar is absent. Adapter, FiLM [17], and attention operations then integrate the controlled state with visual and textual features.

The evaluation focuses on both classification performance and mechanism behavior. The fixed test partition is treated as an internal benchmark split rather than an untouched blind test because its aggregate results informed more than one research decision. Five modality-availability configurations and six fusion families are each trained with seeds 42, 43, and 44 under the same schedule and checkpoint-selection protocol. Visual-only and visual-sensor-state models reach 88.98% and 88.50% mean Accuracy, locating the clean-input discriminative core in the image stream. The full model achieves $88.26\% \pm 0.21\%$ Accuracy and $82.26\% \pm 0.31\%$ Macro F1, improving on QMF by 3.47 and 3.89 percentage points, respectively. Among the three-modality fusion families, LMF reaches comparable Accuracy at $88.30\% \pm 0.18\%$ and the highest clean-input Macro F1 at $83.28\% \pm 1.36\%$. The proposed model instead exhibits lower NLL, Brier score, ECE, and Macro-F1 variation than LMF. Under low-light stress, it exceeds Visual-only by 1.54 points Accuracy and 2.03 points Macro F1. A same-checkpoint counterfactual further reduces Macro F1 from 27.06% with aligned sidecars to 24.21% when missing and 26.21% when globally permuted. Removing the compound reliability/fallback subsystem reduces Accuracy and Macro F1 by mean values of 0.81 and 0.51 points after independent retraining. A matched constant-gate control does not establish a consistent average classification benefit from

sample adaptivity: the dynamic model gains 0.48/0.98 Accuracy/Macro-F1 points under strong Gaussian noise but changes by -0.20/-0.17 points on clean input and -0.19/-0.21 points under mixed faults. Direct gate-score distributions therefore support fault-sensitive ordering, while the compound fallback subsystem carries the principal classification-robustness result.

The contributions of this work are threefold:

1. We formulate reliability-aware waste classification from RGB, category-text prototypes, and a normalized 4×4 sensor-state sidecar, using cross-modal context to predict a continuous fallback for missing records.
2. We develop a sample-wise reliability gate that combines availability, raw-grid statistics, predicted-observed consistency, latent discrepancy, and a bounded variance diagnostic to control the residual contribution of a degraded auxiliary state.
3. We evaluate five independently trained modality configurations and the complete system against five same-input fusion alternatives under a uniform three-seed protocol, together with matched retraining, a same-checkpoint low-light sidecar counterfactual, direct gate-score fault detection, controlled input stress, and a class-memory probe with a nonsemantic class-ID control.

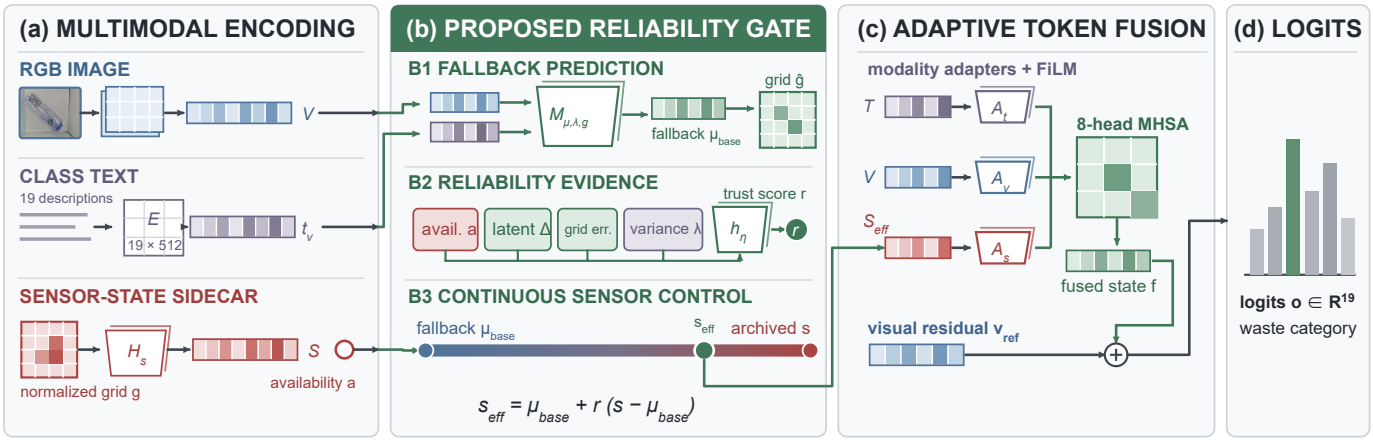


Figure 1. Architecture of the proposed reliability-gated multimodal classifier. Three aligned encoder lanes produce visual, class-text context, and archived sensor-state representations; the sensor lane receives a normalized single-channel 4-by-4 sidecar. Visual and class-text context predict a latent fallback, bounded variance diagnostic, and reference grid. The reliability module estimates sample-wise record trust and controls the archived residual before token adaptation, attention fusion, and classification.

II. RELATED WORK

A. Visual Waste Recognition

Waste recognition spans object classification, litter detection, and industrial instance segmentation. WasteNet provides a compact smart-bin classifier [1]; TACO supplies context-rich litter annotations [2]; and ZeroWaste targets deformable objects under industrial clutter [3]. SortWaste and ScatSpotter further expose dense annotation, camouflage, and collection-distribution effects [4] [5].

Together, these studies show that performance depends on the unit of analysis and evaluation provenance. We therefore focus on object-crop classification and make robustness to sensor quality the central methodological question, with detection and segmentation results serving as adjacent context.

B. Multimodal Sensing for Waste Sorting

Material recognition can benefit from signals that are difficult to infer from visible appearance alone. Bihler *et al.* studied deep fusion of multiple physical sensing channels for bulky-waste classification, including optical and non-visible measurements [6]. SpectralWaste couples RGB and hyperspectral imagery in a sorting-facility setting, supporting multimodal segmentation and sensor-aware automation research [7]. These systems demonstrate the value of physically distinct modalities while also making calibration, spatial alignment, synchronization, and sensor health part of the learning problem.

Our setting uses RGB, class-level text prototypes, and a compact 4×4 normalized sensor-state sidecar. The array preserves its two-dimensional layout instead of flattening the 16 values into unrelated scalar inputs. Unlike fixed-trust fusion, the proposed model tests the current sidecar against a cross-modal prediction before deciding how strongly its residual should influence classification. Independent physical sensing remains a separate transfer setting not established by the present archive.

The empirical comparison spans complementary fusion families rather than a single competing operator. Feature concatenation followed by an MLP represents conventional joint-representation learning [18]. Gated Multimodal Units (GMU) learn multiplicative modality gates [19], whereas Low-rank Multimodal Fusion (LMF) factorizes multiplicative cross-modal interactions [20]. QMF and TMC then provide quality-aware and evidential decision-fusion references, respectively [11] [21]. In our experiments, every method receives the same visual, candidate-prototype, and sensor branches; only the fusion and associated objective are changed.

C. Missing and Degraded Sensor Learning

Missing-modality research has shown substantial vulnerability across multimodal transformer architectures and tasks under absent inputs [8]. Modality-robustness diagnostics further reveal sensitivity to dominant channels [9], while gradient-guided decoupling explicitly reduces conflicting modality dependence when inputs are incomplete [22]. Quality-aware dynamic fusion estimates the contribution of each modality instead of assigning fixed trust; QMF provides a representative formulation for low-quality multimodal data [11]. Trusted multi-view classification models evidential uncertainty across views [21]. Our formulation complements these approaches with a continuous residual path between a predicted sensor state and an archived spatial state.

A second line learns representations that remain usable when one or more channels are unavailable. SMIL uses Bayesian meta-learning under severely incomplete training and test modalities [10], whereas MissModal aligns incomplete and complete representations without changing the downstream fusion operator [23]. Distribution-consistent recovery aligns reconstructed and observed representations [12]; probabilistic conformal distillation incorporates uncertainty [13]; and hyper-modality enhancement builds shared representations [24]. Our method anchors sensor-state fallback prediction to visual features and category-text

prototypes, then scales the archived residual with a sample-wise reliability score rather than separating recovery from fusion.

A third line uses predicted variance as a quality signal. Heteroscedastic modeling distinguishes data-dependent noise from model uncertainty [25], and robust-fusion work uses unimodal aleatoric uncertainty to adjust modality contributions [14]. Calibration studies show that confidence behavior varies with distribution shift, architecture, and pretraining [26] [27] [28]. We therefore use predicted variance as one bounded diagnostic within the reliability gate, alongside availability, raw-grid statistics, predicted-observed disagreement, and latent discrepancy.

The closest prior themes provide quality weighting, missing-representation recovery, and uncertainty estimation. Our contribution is their joint organization around a structured spatial sidecar, an explicit residual-gating equation, and a predefined fallback path. Controlled missing, noise, bias/drift, stuck, and mixed conditions provide a reproducible mechanism profile across complementary fault patterns.

III. METHOD

A. Problem Formulation

For each object, the classifier receives an RGB crop $x \in \mathbb{R}^{3\times480\times480}$, a bank of $K = 19$ class descriptions, a sample-aligned normalized sensor-state grid $g \in [0,1]^{1\times4\times4}$, and an availability indicator $a \in [0,1]$. The class descriptions are encoded once per forward pass as a prototype matrix $E = [e_1, ..., e_K]^\top \in \mathbb{R}^{K\times512}$. The model returns logits $o \in \mathbb{R}^K$.

The RGB crop and sensor-state grid are sample-indexed inputs, whereas E is a fixed closed-set class-text prior rather than a third sample observation. The methodological question is how a classifier can use, attenuate, or replace the auxiliary grid when it is internally consistent, degraded, or missing. The proposed model answers this question through a learned cross-modal fallback state and a sample-wise reliability gate. Its information flow is continuous from (x, E, g, a) to o, as defined below and summarized in Fig. 1.

During training, g^0 denotes the recorded sensor grid before fault injection, and $\tilde{g}$ denotes its optionally degraded version after clipping to $[0,1]$. At clean validation or inference, $\tilde{g} = g^0$ unless an evaluation condition explicitly alters the record. To simplify notation, the forward definition below uses $g = \tilde{g}$.

The implemented record contract is narrower than the sensor hardware contemplated in the originating patent. Each manifest row is associated through 'sample_id' with one pre-normalized 4×4 field stored under 'normalized_grid'; the classifier does not receive a device identifier, physical unit, sampling frequency, power, site spacing, phase, or raw calibration references. The current archive metadata identifies these fields as image-conditioned sidecar estimates, with deterministic fallback for structurally invalid generated records, rather than independently audited hardware measurements. The model therefore treats the 16 ordered values as a sensor-state interface and tests robustness to controlled changes of that interface. Filtering, CLAHE, sharpening, and affine correction are applied to the RGB crop before resizing to 480×480; no depth tensor is passed to the network or counted as an additional modality.

B. Multiscale Visual Representation and Class-Text Prototypes

A YOLO11s-seg backbone [15] supplies the three feature maps used by its segmentation head. The implementation reads the source indices from the head's `f` field at runtime; for the archived `yolo11s-seg.pt` checkpoint they are layers 16, 19, and 22. For scale $k \in \{1,2,3\}$, global average pooling and a lazy linear projection first unify the backbone output dimension, after which a fusion-layer projection produces

$$\begin{aligned} h_k &= Q_k\big(\ \mathrm{GAP}\ (F_k(x))\ \big) \in \mathbb{R}^{512}, \\ z_k &= P_k(h_k) \in \mathbb{R}^{512}. \end{aligned} \qquad (1).$$

The initial visual state is

$$\alpha_k^{(0)} = \frac{\exp\,(p_k)}{\sum_{j=1}^{3}\exp\,(p_j)}, \qquad v = \sum_{k=1}^{3}\alpha_k^{(0)} z_k \qquad (2),$$

where the learnable logits p are initialized from the normalized prior $(0.3, 0.5, 0.2)$. Each $P_k: \mathbb{R}^{512} \to \mathbb{R}^{512}$ is initialized as an identity map; Q_k resolves its input width on the first forward pass. The first five YOLO model layers are frozen; the remaining participating layers are trainable.

Each of the 19 classes uses one fixed Simplified-Chinese UTF-8 description string without a prefix or suffix template. The strings are batched in manifest order, tokenized with padding and truncation to CLIP's 77-token maximum, and encoded by the CLIP ViT-B/16 text tower [16]. The exact strings, human-readable category names, manifest order, and human-verified English explanations are listed in the anonymous supplement. The first three CLIP text-transformer layers are frozen. The projected text embeddings are L_2-normalized to form $E \in \mathbb{R}^{19\times512}$. No ground-truth label is supplied to this branch. Instead, the visual state queries every class prototype:

$$\pi^v = \mathrm{softmax}\left(\frac{\bar{v}\,\bar{E}^\top}{\tau}\right), \qquad t_v = \pi^v E, \qquad \tau = 0.07 \qquad (3),$$

where bars denote row-wise L_2 normalization. Thus, $t_v \in \mathbb{R}^{512}$ is a soft semantic context over all candidate classes, not the embedding of the target label.

C. Sensor Encoding and Cross-Modal Fallback Prediction

The current sidecar stores the network-ready field directly. For row i and column j, the loader uses

$$g_{ij}^0 = [\texttt{normalized_grid}]_{ij} \in [0,1],$$

and verifies that the nested record is strictly 4×4 and finite. The archive also contains five scalar sidecar attributes named weight, temperature, moisture, conductivity, and odor response. They are neither grid channels nor classifier inputs. Because the current records do not include raw device units or per-site calibration references, no hardware calibration equation is inferred from the normalized values.

The observed sensor grid is standardized per sample before latent encoding:

$$g_{std} = \frac{g - \ \mathrm{mean}\ (g)}{\max\,(\ \mathrm{std}\ (g), 10^{-6})} \qquad (4).$$

The implementation uses the population standard deviation. Following the configured $4\times4 \to 16\times16$ preprocessing path, it bilinearly upsamples g_{std}, applies a 1×1 convolution from 1 to 512 channels, a LeakyReLU with slope 0.1, and global average pooling:

$$s = H_s(g) \in \mathbb{R}^{512} \qquad (5).$$

Standardization intentionally removes sample-wise offset and scale from the latent path. The unstandardized grid remains available to the quality heads below, so a constant or saturated grid is not hidden by this normalization.

The visual state and early class-text context predict a fallback sensor latent mean and a bounded log variance:

$$\begin{aligned} c &= [v; t_v] \in \mathbb{R}^{1024}, \\ \mu_{base} &= M_\mu(c) \in \mathbb{R}^{512}, \\ \lambda &= l_{\min} + (l_{\max} - l_{\min})\ \mathrm{sigmoid}\ (M_\lambda(c)), \end{aligned} \qquad (6)$$

with $l_{\min} = -3$ and $l_{\max} = 2$. Both M_μ and M_λ have the sequence $1024 \to 512$, LeakyReLU(0.1), dropout(0.3), and $512 \to 512$. Here $\lambda = \log \sigma^2$ is a variance-shaped latent diagnostic inspired by heteroscedastic modeling [25]. Its role is bounded error ranking within the sensor-reliability head; an absolute physical calibration is not inferred from the archived normalized values.

The same context predicts a clean spatial sensor grid:

$$\widehat{g} = \text{reshape}_{1\times 4\times 4}\left[\text{sigmoid}\left(M_g([v; t_v; \mu_{base}])\right)\right] \quad (7),$$

where M_g is a $1536 \to 64 \to 16$ multilayer perceptron with LeakyReLU(0.1) between the two linear layers. Unlike s, $\widehat{g}$ preserves the explicit 4×4 spatial interface and is trained against g^0.

D. Sensor-Reliability Evidence

Latent agreement between the observed representation and its predicted fallback is summarized by

$$\begin{aligned} D &= \text{clip}_{[0,10^6]}\left(\tfrac{1}{512}\textstyle\sum_{d=1}^{512}(s_d - \mu_{base,d})^2 e^{-\lambda_d}\right), \\ U &= \text{clip}_{[0,10^6]}\left(\tfrac{1}{512}\textstyle\sum_{d=1}^{512} e^{\lambda_d}\right), \\ C &= \text{clip}_{[0,1]}\left(\tfrac{\cos(s,\mu_{base})+1}{2}\right). \end{aligned} \quad (8)$$

The latent head receives

$$e_{lat} = [\log(1+D),\ C,\ \log(1+U),\ a] \in \mathbb{R}^4 \quad (9).$$

Raw-grid evidence $e_{raw} \in \mathbb{R}^7$ is computed directly from g: mean, population standard deviation, dynamic range, mean absolute horizontal gradient, mean absolute vertical gradient, saturation ratio at values ≤ 0.02 or ≥ 0.98, and flatness $\exp(-10\ \text{std}(g))$.

Predicted-grid evidence compares the archived observation with the cross-modal sensor-state prediction. Let $Q(\cdot)$ denote the seven raw statistics. The resulting 22-dimensional vector is

$$\begin{aligned} e_d &= [\text{MAE}, \text{RMSE}, \text{MaxAE}, \\ &\quad |\text{mean}(g - \text{sg}(\widehat{g}))|, \cos(g, \text{sg}(\widehat{g}))], \\ e_\nabla &= [\Delta_h, \Delta_v], \\ e_{cross} &= [Q(g), Q(\text{sg}(\widehat{g})), e_d, e_\nabla, a], \end{aligned} \quad (10)$$

where $e_d \in \mathbb{R}^5$ contains cell-level discrepancy measures and Δ_h and Δ_v are the mean absolute differences between observed and predicted horizontal and vertical finite differences. Thus, $e_{cross} \in \mathbb{R}^{22}$. The operator sg denotes stop-gradient. In code, the entire predicted grid is detached before its statistics and comparison terms are formed. This assigns grid prediction to clean-grid reconstruction and reliability estimation to observed-predicted comparison.

Three scalar-logit heads aggregate the evidence:

$$\begin{aligned} \eta &= h_{lat}(e_{lat}) + h_{raw}(e_{raw}) + h_{cross}(e_{cross}), \\ r &= a\ \text{sigmoid}(\eta). \end{aligned} \quad (11).$$

Their dimensions are $4 \to 32 \to 1$, $7 \to 16 \to 1$, and $22 \to 32 \to 1$, respectively, each with LeakyReLU(0.1). The latent reliability head is initialized to $r = 0.85$ before availability multiplication; the raw- and predicted-grid output layers are initialized to zero, so they initially contribute neutral logit residuals.

The effective sensor state is the predicted fallback plus a gated observed residual:

$$s_{eff} = \mu_{base} + r(s - \mu_{base}) \quad (12).$$

Equation (12) gives an exact missing-data path: $a = 0 \Rightarrow r = 0 \Rightarrow s_{eff} = \mu_{base}$. For an available record, r continuously interpolates between the fallback and observation. Operationally, r estimates agreement with the archived sensor-state grid under the defined fault-injection protocol. It is not asserted to be a device-health probability or a direct estimate of sample-level classification benefit.

The matched gate-adaptivity control replaces the sample-wise logit η only at the final step by the reliability head's learned output bias b_r, giving $r_{const} = a\ \text{sigmoid}(b_r)$. It retains the same fallback, availability rule, fusion paths, parameter inventory, initialization, losses, and optimization schedule. Thus, the control removes evidence-dependent variation in r without removing the fallback subsystem or changing the exact missing-input endpoint.

E. Interpolation Properties and Design Rationale

The residual form in (12) imposes several useful structural constraints before any parameter is learned. Since $a \in [0,1]$ and the sigmoid output lies in $[0,1]$, the effective coefficient satisfies $0 \le r \le a \le 1$. Consequently, s_{eff} lies on the line segment between μ_{base} and s, and $\| s_{eff} - \mu_{base} \|_2 = r \| s - \mu_{base} \|_2$. The observed latent residual is therefore bounded by its original magnitude, giving the learned fusion path an interpretable local geometry.

Three endpoint cases follow directly. First, an unavailable record has $a = 0$, which forces exact fallback regardless of the learned reliability logit. Second, an available but rejected observation has $r \approx 0$ and is treated like a missing record. Third, a strongly trusted observation has $r \approx 1$, for which the effective state approaches the encoded archived state. Unlike a hard switch, intermediate values retain a differentiable path and allow evidence to be partially discounted.

Figure 2 consolidates the reliability mechanism at three complementary levels. Panel (a) traces the availability, latent, raw-grid, and predicted-grid evidence that produces the sample-wise trust score. Panel (b) shows the bounded latent interpolation and its exact missing-input boundary condition. Panel (c) connects the same coefficient to the missing, degraded, and strongly trusted operating regimes.

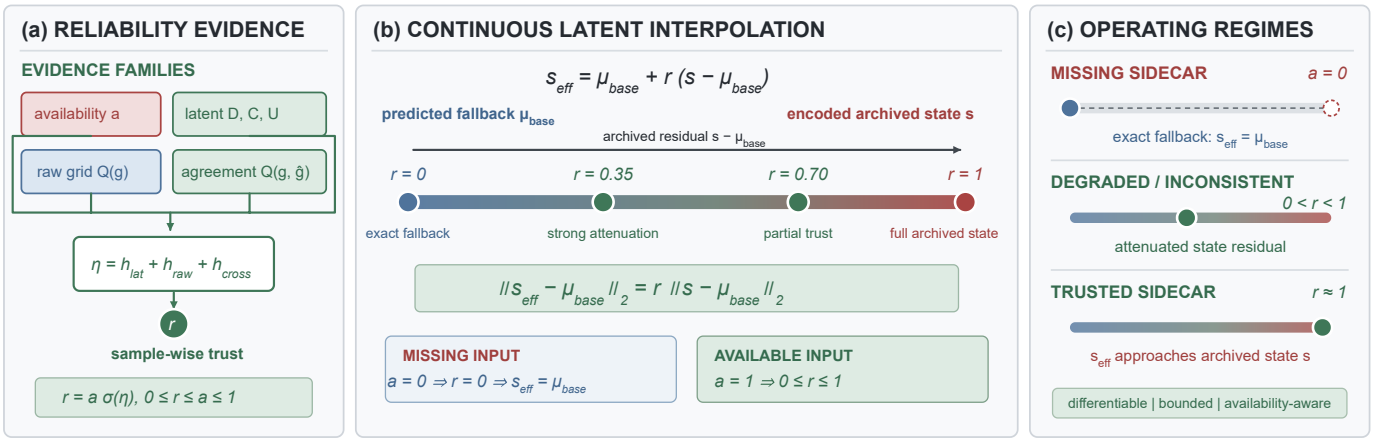


Figure 2. Reliability evidence, continuous latent interpolation, and operating regimes. The effective sensor state remains on the line segment between the learned fallback and the encoded archived state; availability forces the missing-input endpoint, while learned evidence continuously attenuates an available record.

The gate is deliberately supported by nonredundant evidence families. Availability provides a deterministic boundary condition; raw-grid statistics expose saturation, flatness, and local discontinuity; predicted-grid comparisons test whether spatial structure is plausible under the learned cross-modal prior; latent discrepancy evaluates agreement after sensor encoding; and predicted variance supplies a learned error-ranking cue. Table I summarizes these roles. Their combination addresses cases in which smoothness, visual plausibility, or latent agreement alone would be ambiguous.

TABLE I. RELIABILITY EVIDENCE USED BY THE SENSOR GATE AND ITS INTENDED DIAGNOSTIC ROLE.

Evidence family	Representative inputs	Primary failure cue	Role in the gate
Availability	a	Missing sidecar record	Forces exact fallback when zero
Raw spatial statistics	Mean, spread, range, gradients, saturation, flatness	Clipping, low dynamic range, local discontinuity	Describes the observation without semantic assumptions
Predicted-grid agreement	Cell error, gradient error, cosine agreement	Structure inconsistent with cross-modal context	Tests spatial plausibility against $\hat{g}$
Latent agreement	Mahalanobis-like discrepancy and cosine similarity	Encoded observation far from μ_{base}	Compares observed and fallback representations
Variance diagnostic	Mean $\exp(\lambda)$	Difficult cross-modal sensor prediction	Supplies a bounded relative uncertainty cue

Stop-gradient in the predicted-grid comparison separates two responsibilities: clean-grid reconstruction trains the fallback predictor, while reliability supervision trains the gate to recognize agreement with the recorded pre-injection grid. This blocks a direct shortcut between the quality objective and its own comparison target and keeps the two training signals interpretable. Section IV-H evaluates this routing choice through a matched three-seed control that allows reliability gradients to propagate into the predicted-grid decoder.

F. Adapter, FiLM, Attention, and Residual Classification

The final semantic context is queried after sensor-reliability control:

$$\pi = \mathrm{softmax}\left(\frac{(v + \bar{s}_{eff})\,\bar{E}^{\top}}{\tau}\right), \qquad t = \pi E \qquad (13).$$

Let $h_1 = v$, $h_2 = t$, and $h_3 = s_{eff}$. Each branch has a separate adapter $A_i : \mathbb{R}^{512} \to \mathbb{R}^{512}$ followed by LeakyReLU(0.1) and dropout(0.3). Global trainable FiLM-style parameters [17], rather than sample-conditioned generators, transform the tokens:

$$u_i = \gamma_i \odot A_i(h_i) + \beta_i \qquad (14),$$

where $\gamma, \beta \in \mathbb{R}^{3\times 512}$ are initialized to 1.0 and 0.1. For each token, its mean cosine agreement with the other two tokens defines

$$b_i = \frac{1}{2}\sum_{j \neq i} \cos(u_i, u_j), \qquad \rho = \mathrm{softmax}(b) \qquad (15).$$

The weighted tokens are projected to 128 dimensions, $d_i = W_{\downarrow}(\rho_i u_i)$, and passed through eight-head self-attention over the three-token sequence:

$$A = \mathrm{MHA}_8(D, D, D), \quad u_i' = u_i + W_{\uparrow} A_i, \quad f = \sum_{i=1}^{3} \rho_i u_i'. \qquad (16).$$

Both $W_{\downarrow}$ and $W_{\uparrow}$ are learned linear maps, $512 \to 128$ and $128 \to 512$. The complete configuration enables all three branches, so no modality mask is active.

Finally, f refines the original visual-scale weighting:

$$\alpha = \mathrm{softmax}(p + G_{scale}(f)), \qquad v_{ref} = \sum_{k=1}^{3} \alpha_k z_k \qquad (17),$$

where $G_{scale} : \mathbb{R}^{512} \to \mathbb{R}^3$ is initialized to zero. The continuous forward map terminates at

$$o = W_c\ \mathrm{LayerNorm}(f + v_{ref}) + b_c, \qquad o \in \mathbb{R}^{19} \qquad (18).$$

G. Reliability Supervision and Joint Objective

Training corruptions are applied to g^0 before per-sample standardization. The configured probabilities are 0.10 for missing, 0.50 for Gaussian noise, 0.25 for directional bias/drift, and 0.15 for a stuck grid. Missing records are zeroed and excluded from subsequent corruptions. Noise standard deviation is sampled from $[0.02, 0.15]$, bias magnitude from $[0.075, 0.15]$, and the final grid is clipped to $[0, 1]$.

The supervision target

$$y_{rel} = a \cdot \mathrm{clip}_{[0,1]}\left(1 - \frac{\mathrm{mean}\,|g - g^0|}{0.25}\right) \qquad (19)$$

measures agreement with the archived grid before fault injection. A high y_{rel} states that the current input is close to g^0

under the chosen L_1 scale. Classification benefit and probability calibration are evaluated independently through the task metrics.

Let p_y be the softmax probability of the correct class and let $s^0 = H_s(g^0)$. The classification loss is

$$\mathcal{L}_{cls} = \mathcal{L}_{CE} - \frac{1}{B}\sum_{i=1}^{B} 0.25(1-p_{y_i})^2 \log p_{y_i}, \qquad (20)$$

which combines unit-weight cross-entropy and multiclass focal loss [29]. The auxiliary objectives are

$$\begin{aligned} \mathcal{L}_{rec} &= \text{mean}_{i,d}\ \tfrac{1}{2}\Big[(\mu_{base,id} - \text{sg}(s_{id}^0))^2 e^{-\lambda_{id}} + \lambda_{id}\Big], \\ \mathcal{L}_{cal} &= \frac{\sum_i \omega_i\ \text{BCE}(r_i, y_{rel,i})}{\sum_i \omega_i}, \\ \mathcal{L}_{var} &= \text{mean}_{i,d}\ (\lambda_{id}+2)^2, \\ \mathcal{L}_{grid} &= \text{SmoothL1}(\widehat{g}, \text{sg}(g^0)). \end{aligned} \qquad (21)$$

The calibration weight is $\omega_i = 3$ when $y_{rel,i} = 1$ and $\omega_i = 1$ otherwise, emphasizing clean records without widening the displayed equation. Although model output already bounds λ to $[-3, 2]$, the reconstruction implementation additionally clips it to $[-12, 12]$ before exponentiation. The clean latent target is computed under `no_grad`, and both clean targets are detached. The reconstruction mask is all ones, so missing observations retain fallback and clean-grid supervision.

For batch pairs $\mathcal{P} = \{(i,j): i < j,\ |y_{rel,i} - y_{rel,j}| \geq 0.05\}$, define $\ell_i = \text{logit}(\text{clip}(r_i, 10^{-6}, 1-10^{-6}))$. Then

$$\begin{aligned} \mathcal{L}_{rank} &= \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \\ &\text{BCEWithLogits}(\ell_i - \ell_j, 1[y_{rel,i} > y_{rel,j}]). \end{aligned} \qquad (22)$$

with zero loss when $\mathcal{P}$ is empty. Define the auxiliary-loss and weight vectors as $\ell_{aux} = [\mathcal{L}_{rec}, \mathcal{L}_{cal}, \mathcal{L}_{var}, \mathcal{L}_{grid}, \mathcal{L}_{rank}]^\top$ and $\mathbf{w} = [0.1, 0.3, 0.1, 0.5, 0.05]^\top$. The full objective is

$$\mathcal{L}_{main} = \mathcal{L}_{cls} + \mathbf{w}^\top \ell_{aux} \qquad (23).$$

Equation (23) is the complete objective of the proposed 100-epoch model used in the principal comparisons, robustness study, ablations, and gate diagnostics. It jointly trains visual classification, the cross-modal fallback μ_{base}, and reliability-gated fusion for RGB, class-text prototypes, and the normalized 4×4 sensor-state sidecar.

H. Developmental Class-Memory Fallback Probe

Section IV-J studies a separate mechanism probe built after the principal model was fixed. It is not the proposed model reported in the main comparison, robustness, ablation, or gate-analysis tables. The probe freezes a second CLIP text tower, maps each class-aligned prompt d_c to a normalized key k_c, and aligns a visual query with the key bank:

$$\begin{aligned} q_{sem} &= \text{norm}(W_q v), \qquad K_{sem} = [k_1, \ldots, k_K]^\top, \\ \pi^{sem} &= \text{softmax}(q_{sem} K_{sem}^\top / \tau). \end{aligned}$$

The retrieval weights address a learned grid-value bank $B = [b_1, \ldots, b_K]^\top$ and fixed training-split sensor centers $C = [m_1, \ldots, m_K]^\top$:

$$u_{sem} = \pi^{sem} B, \qquad m_c = \frac{1}{N_c}\sum_{i:\, y_i = c} H_s(g_i^0), \qquad \mu_{sem} = \sum_c \pi_c^{sem} m_c$$

Only in this probe, μ_{sem} replaces μ_{base} as the endpoint in (12), while u_{sem} conditions the grid decoder. Its query loss is $\mathcal{L}_{sem} = \text{CE}(q_{sem} K_{sem}^\top / \tau, y)$, and the refinement objective is $\mathcal{L}_{probe} = \mathcal{L}_{main} + \lambda_{sem}\mathcal{L}_{sem}$ with $\lambda_{sem} = 0.2$. Ground-truth labels supervise the alignment loss only during refinement; inference queries all K keys without a target label. A protocol-matched nonsemantic control replaces K_{sem} with the first K rows of the 512-dimensional identity matrix. These fixed orthogonal class-ID keys preserve one-to-one class-memory addressing and capacity while removing language content. This notation and separation prevent the developmental class-memory result from being conflated with the 100-epoch proposed architecture.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Evaluation Protocol

We evaluate object-level classification on an internal 19-class waste dataset containing 8,654 annotated objects from 5,990 RGB source images. The RGB files are associated with LabelMe polygon annotations [30], and each object record is joined by sample identifier to a normalized 4×4 sensor-state sidecar. The archive preserves object-level pairing, array topology, and source-image groups for evaluation.

The split was generated by a stable hash of `source_image_id`, keeping all objects from one image in the same partition. Table II summarizes the resulting 7,002/826/826 objects and 4,823/597/570 source images for training, validation, and testing. No byte-identical images cross partitions. A conservative near-duplicate screen identified nine affected test objects; excluding them changed the seed-42 Accuracy/Macro F1 only from 88.14/82.44% to 88.00/82.37%, confirming that the aggregate result is stable to those cases.

TABLE II. INTERNAL BENCHMARK PARTITIONS AFTER SOURCE-IMAGE-GROUPED SPLITTING.

Partition	Objects	Source images	Classes	Object share (%)
Training	7,002	4,823	19	80.91
Validation	826	597	19	9.54
Internal test	826	570	19	9.54
Total	8,654	5,990	19	100.00

All 8,654 object records contain a sample-indexed 4×4 field under `normalized_grid`. Current provenance metadata identifies 8,639 records as image-conditioned vision-language estimates and 15 as deterministic fallbacks after structured-output failure. The same JSONL rows contain five scalar sidecar attributes named weight, temperature, moisture, conductivity, and odor response; none is passed to the classifier. The loader consumes only the normalized grid, whose stored values are finite and bounded to $[0, 1]$. The archive does not include a device identifier, physical unit, sampling frequency, site spacing, or raw calibration references, and the experiment therefore does not characterize an independently measured hardware modality.

The test split contributes neither gradients nor checkpoint selection. Because aggregate test performance informed later model comparisons, we treat it as a held-out internal benchmark rather than an untouched external blind test.

B. Sensor-State Record Interface and Controlled Faults

The multimodal join is object based. Each object manifest row contains a unique `sample_id`, a source-image group identifier, a bounding box, and one of the 19 class labels; the sidecar table is keyed by the same `sample_id` and supplies exactly 16 normalized values in four ordered rows. This one-to-one key prevents the loader from assigning a grid by filename order or batch position.

Multiple objects may originate from one RGB image, but they remain distinct object-sidecar pairs and are kept in the same partition through the source-image group.

The archived grid is already bounded to $[0, 1]$; the current loader performs shape and finiteness validation rather than reconstructing an undocumented hardware calibration. Equation (4) standardizes the 16 values only inside the sensor-state latent branch. Their original row-column order is retained through the $4 \times 4 \rightarrow 16 \times 16$ interpolation. The reliability heads receive the unstandardized bounded grid, so they can still observe mean, variance, dynamic range, saturation, and flatness. This separation lets the latent encoder emphasize spatial response shape while the quality path retains the absolute cues needed to recognize clipping or a constant record.

The originating patent describes an RGB-D camera and a 4×4 sensor array, but the experimental input contract available here is specifically a three-channel RGB crop plus the normalized sidecar; no depth tensor or hardware acquisition metadata is present.

Figure 3 separates the implemented record contract into three auditable stages. Panel (a) extracts the annotated RGB object crop. Panel (b) validates the image-conditioned 4×4 normalized sidecar and keeps its five scalar attributes outside the classifier. Panel (c) joins the two neural inputs by `sample_id` and applies `source_image_id` only as the group key for leakage-free partitioning. Table III fixes the exact archived-record and neural-input boundary without inferring unavailable hardware parameters.

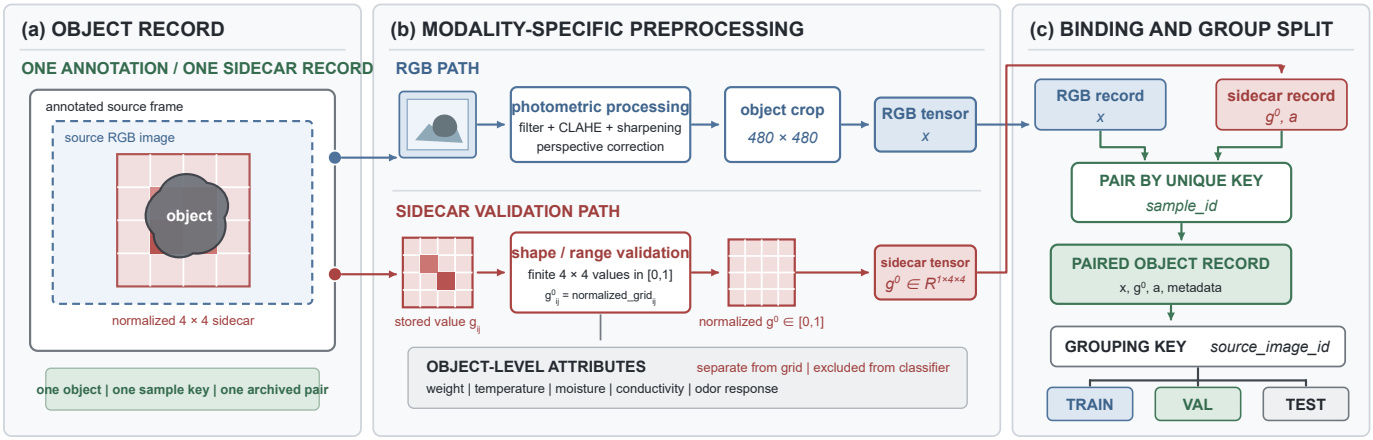


Figure 3. Object-record construction and grouped sample binding. An annotated RGB crop and its image-conditioned normalized 4×4 sidecar are joined by sample identifier; five scalar sidecar attributes are archived separately and excluded from the classifier.

TABLE III. ARCHIVED SIDECAR FIELDS AND THE EXACT STORED-INPUT BOUNDARY USED BY THE EXPERIMENTS.

Record field	Archived benchmark and model contract
Provenance	8,639 image-conditioned vision-language estimates; 15 deterministic fallbacks
Spatial topology	4×4 ordered values; 16 positions per object record
Stored source field	`normalized_grid`; finite nested array with exact shape 4×4
Channel organization	One stored spatial channel
RGB-sidecar association	Joined one-to-one by `sample_id`
Spatial indexing	Four retained row-column coordinates; not flattened before tensor construction
Stored network value	$g^0_{ij} = [\texttt{normalized_grid}]_{ij}$
Stored network range	$g^0_{ij} \in [0, 1]$
Neural sensor tensor	$1 \times 4 \times 4$, bilinearly expanded to 16×16, then 1×1 projected from 1 to 512 channels
Camera tensor	RGB crop only, resized to $3 \times 480 \times 480$; depth is excluded
Excluded scalar sidecars	Weight, temperature, moisture, conductivity, and odor response; not classifier inputs
Unavailable hardware fields	Device model, physical unit, frequency, power, site spacing, dwell time, and raw calibration references

The test sidecars span both spatially uniform and strongly localized patterns. Of 826 grids, 309 (37.4%) are exactly flat after stored normalization; 311 (37.7%) have dynamic range at most 0.05. Across all records, dynamic range has mean 0.284 and median 0.170, spatial variance has mean 0.0234 and median 0.0025, and the mean neighboring-cell gradient has mean 0.0899 and median 0.0300. Exact flatness denotes the absence of within-grid contrast, not a missing record: the unstandardized quality path retains its absolute level, while the standardized latent branch maps a constant field to zero shape contrast. Figure 4 shows representative fields, the full range distribution, and category-wise median/interquartile range using human-readable object names.

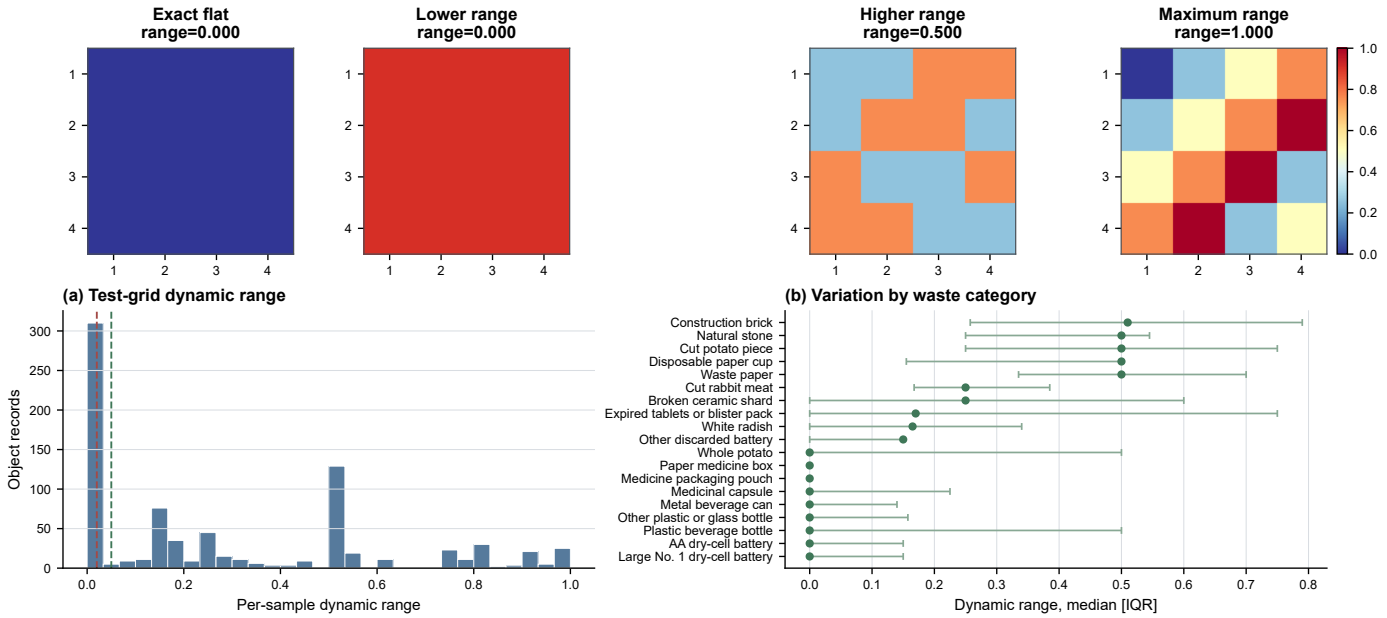


Figure 4. Spatial information in the archived test arrays. The upper row shows representative normalized fields from exact-flat to maximum-range records. The lower panels report the per-object dynamic-range distribution and category-wise median with interquartile range.

Controlled faults are deterministic given the perturbation seed. Complete absence sets the availability indicator to zero and activates the exact fallback in (12). Gaussian noise perturbs individual cells before clipping; directional bias changes the spatial response coherently; the stuck condition collapses the usable variation; and the mixed condition composes the configured training-time fault process. For a paired comparison, the full and ablated models receive identical perturbed records. The five perturbation seeds therefore measure sensitivity to fault realization, while model-training variability is reported separately with seeds 42, 43, and 44.

These transformations are software interventions on archived sidecars. They provide controlled counterfactual inputs that isolate gate behavior across reproducible changes in availability, noise, bias, and spatial pattern.

C. Implementation and Metrics

The model was implemented with Python 3.9.23 and PyTorch 2.6.0 on one NVIDIA RTX 4090 (24 GB). RGB inputs were resized to 480×480. The visual branch was initialized from YOLO11s-seg [15], and category prototypes were encoded with CLIP ViT-B/16 [16]. The visual, text, and sensor-state adapters use 512-dimensional representations; fusion uses 128-dimensional tokens and eight-head self-attention. Dropout was 0.3.

All methods used AdamW [31], batch size 16, an initial learning rate of 10^{-4}, weight decay 10^{-5}, five linear warm-up epochs, cosine decay, and 100 epochs. The proposed-model objective combined cross-entropy and focal loss [29] with sensor-latent reconstruction, reliability calibration, variance-prior, grid-reconstruction, and reliability-ranking terms; the corresponding auxiliary weights were 0.1, 0.3, 0.1, 0.5, and 0.05. Concat-MLP, GMU, and the principal LMF reference used cross-entropy, while QMF and TMC retained their adapted quality-ranking and evidential objectives. A second LMF control used the same unit-weight CE-plus-focal classification objective as the modality matrix and proposed classifier, without adding the proposed auxiliary terms. Best checkpoints were selected only by validation Accuracy. Every entry in the main comparison was independently trained with seeds 42, 43, and 44. For the proposed model, the selected epochs were 92, 91, and 92.

Training-time fault injection was applied before grid standardization. A record was made unavailable with probability 0.10. Otherwise, Gaussian cell noise, directional bias, and a stuck-grid operation were applied with probabilities 0.50, 0.25, and 0.15, respectively. Clean validation and clean test evaluation used the recorded sensor grids without random degradation.

The comparison addresses both modality availability and fusion design. The modality matrix retrains visual-only (V), sensor-only (S), visual-text (V+T), visual-sensor (V+S), and text-sensor (T+S) classifiers. Only the named encoders are active; their adapted 512-dimensional representations are averaged, CE plus focal loss is retained, and reliability auxiliaries and training-time sensor faults are disabled so that the clean information carried by each input set is isolated. A text-only row is not included because the fixed class-description bank contains no sample-specific observation; without a visual or sensor query it produces the same prior for every object.

The three-modality comparison covers early, gated, low-rank multiplicative, quality-aware, and evidential fusion. Concat-MLP concatenates the three 512-dimensional branch features and applies a two-layer classifier [18]. GMU uses modality-specific candidate transforms with feature-wise softmax gates [19]. Rank-4 LMF directly factorizes the three-way interaction into modality-specific factors [20]. The QMF and TMC adaptations retain their quality-weighted and evidential decision rules [11] [21]. All five alternatives use the same encoders, candidate descriptions, normalized sidecars, grouped partitions, preprocessing, training-time grid degradation, 100-epoch schedule, validation-Accuracy checkpoint rule, and three training seeds as the proposed model. The fusion operator and its associated objective are the only method-specific changes.

We report Accuracy, macro-averaged Precision, Recall, and F1, Weighted F1, negative log-likelihood (NLL), multiclass Brier score, and 15-bin expected calibration error (ECE) [26]. NLL and Brier score complement the bin-dependent ECE because predictive uncertainty can behave differently under architecture changes and distribution shift [27] [28]. All macro calculations retain the 19-label space and use `zero_division=0`. This matters because the least frequent test class has only five examples.

Uncertainty intervals respect the data hierarchy. For one trained model, we sample the 570 source-image clusters with replacement

and retain all objects within each sampled cluster, using 2,000 percentile-bootstrap repetitions [32]. Model differences use 5,000 paired cluster-bootstrap repetitions, with identical sampled clusters for both models. Cluster bootstrap is the primary interval procedure; exact McNemar tests provide a complementary object-level sensitivity statistic. Mean and sample standard deviation across the three training seeds summarize observed training variability.

The proposed seed-42 checkpoint is the prespecified reference for detailed class-wise, paired robustness, variance, and timing analyses. All three seeds contribute to the main model comparison, retrained ablations, inference-path interventions, and direct reliability-score diagnostics. This division avoids treating five fault realizations as five additional trained models and keeps post-hoc diagnostics distinguishable from independent retraining.

We additionally evaluate a post-hoc deployment-oriented ensemble without further training. Arithmetic class probabilities from the three completed checkpoints are pooled with nonnegative weights on a simplex grid at increments of 0.05. Weights are selected on validation Macro F1, with Accuracy, NLL, and distance from uniform weights used as deterministic tie-breaks; the resulting weights are then locked before test evaluation. The selected proposed-model weights are 0.30, 0.10, and 0.60 for seeds 42, 43, and 44. For protocol-matched three-checkpoint ensemble comparisons, the identical procedure is independently applied to the three QMF checkpoints and the three matched reliability/fallback ablation checkpoints, selecting weights 0.35/0/0.65 and 0.15/0.30/0.55, respectively. Test labels are not used in any weight search. Because the ensemble protocol was designed after the internal benchmark had already informed the study, it is reported as post-hoc rather than confirmatory.

The dynamic-versus-constant gate control uses the seed-42/43/44 proposed checkpoints and independently trained constant-gate counterparts. The only functional change is r being sample adaptive or equal to the learned scalar r_{const} for every available record; all fallback, fusion, objective, optimizer, epoch, and validation-selection settings are retained. Constant-gate training uses four persistent loader workers for throughput with a separate worker RNG and a sampler that reproduces the archived zero-worker permutation stream. A pretraining audit verifies identical sample-order hashes over two consecutive epochs in every seed, after which the same seeded sampler is used for full training. All clean and fault evaluations use the canonical batch-16, zero-worker protocol.

The class-memory fallback study is a separate developmental mechanism probe and is not the proposed 100-epoch model in Tables IV–IX or the direct gate analysis. For each seed, the final description-key checkpoint initializes from the corresponding eight-epoch query checkpoint and continues for three epochs with AdamW, batch size 16, learning rate 2×10^{-5}, missing probability 0.30, and validation grid-reconstruction error as the checkpoint criterion. The independent text tower and sensor encoder remain frozen, and sensor-state centers are computed only from the 7,002 training records. Four conditions are paired interventions on each description-key checkpoint: vision-conditioned fallback suppresses class-memory retrieval; correct descriptions retain the class-aligned name-description prompts; class-name only replaces each prompt with its category name; and shuffled descriptions cyclically permute complete prompts across the fixed class-bound memory rows. A fifth, separately trained control replaces text-derived keys with fixed orthogonal class-ID vectors under the same seed-specific 8+3-epoch protocol. Missing-100 sets $a = 0$ for every one of the 826 internal-test objects. Reconstruction targets remain the paired clean grid g^0 and its detached sensor encoding $H_s(g^0)$.

Direct gate evaluation uses all three full-model checkpoints without parameter updates. Each checkpoint evaluates the same 826 internal-test objects under clean input, complete absence, Gaussian noise with standard deviations 0.03/0.08/0.15, directional bias with strengths 0.08/0.15, a first-cell latch replicated across the array, and the mixed fault process. Five perturbation seeds (2026–2030) instantiate each stochastic condition. Metrics are first averaged over fault realizations within each trained checkpoint and are then summarized by mean and sample SD across training seeds. For clean-versus-fault discrimination, clean is positive and a record is an effective negative when injection changes its quality target from one. This excludes no-op injections while retaining nominal-condition metrics in the evaluation archive. AUROC is prevalence invariant; AUPRC reweights effective faults to the same total weight as clean objects, giving a 0.5 reference. Complete absence is reported as a deterministic availability boundary and is excluded from the abstract's learned available-fault macro.

D. Main Internal-Benchmark Results

Table IV compares five modality-availability configurations and six three-modality fusion families under one three-seed training and evaluation protocol. Mean proposed-model Accuracy was $88.26\% \pm 0.21\%$, and mean Macro F1 was $82.26\% \pm 0.31\%$. For the prespecified seed-42 reference run, the source-image-cluster 95% intervals were [85.33, 90.79]% for Accuracy and [76.46, 86.22]% for Macro F1. The approximately six-point gap between Accuracy and Macro F1 is consistent with the marked class imbalance: training support ranges from 28 to 833 objects, and a few errors strongly affect the rare classes.

TABLE IV. UNIFORM MODALITY-AVAILABILITY AND FUSION-FAMILY COMPARISON ON THE FIXED INTERNAL BENCHMARK. EVERY ENTRY IS THE MEAN ± SAMPLE SD OVER TRAINING SEEDS 42, 43, AND 44; THE SECOND LMF ROW IS AN OBJECTIVE-MATCHED FAIRNESS CONTROL.

Method	Accuracy (%)	Macro P (%)	Macro R (%)	Macro F1 (%)	Weighted F1 (%)	NLL	Brier	ECE (%)
Visual only (V)	88.98 ± 0.67	82.82 ± 0.48	86.94 ± 1.54	83.95 ± 1.06	89.04 ± 0.68	0.345 ± 0.009	0.159 ± 0.005	2.78 ± 0.48
Sensor only (S)	19.61 ± 1.51	10.21 ± 0.69	12.64 ± 1.25	8.28 ± 1.01	12.63 ± 1.25	2.431 ± 0.024	0.885 ± 0.005	4.13 ± 0.68
Visual + text (V+T)	88.26 ± 0.48	81.93 ± 0.37	84.59 ± 0.64	82.24 ± 0.70	88.31 ± 0.43	0.363 ± 0.011	0.170 ± 0.005	2.63 ± 0.50
Visual + sensor (V+S)	88.50 ± 0.85	82.39 ± 2.80	85.75 ± 1.39	83.32 ± 1.92	88.59 ± 0.76	0.360 ± 0.005	0.168 ± 0.003	2.56 ± 0.26
Text + sensor (T+S)	16.06 ± 0.25	6.02 ± 1.51	9.81 ± 0.14	4.91 ± 0.15	8.17 ± 0.26	2.530 ± 0.003	0.903 ± 0.001	2.54 ± 0.85
Concat-MLP	86.56 ± 1.11	81.21 ± 1.26	81.81 ± 1.28	80.49 ± 1.25	86.51 ± 1.10	0.408 ± 0.011	0.193 ± 0.008	2.67 ± 0.73
GMU	86.84 ± 0.25	81.71 ± 0.13	83.75 ± 1.32	81.93 ± 1.21	86.86 ± 0.45	0.387 ± 0.004	0.185 ± 0.002	2.87 ± 0.61
LMF	88.30 ± 0.18	82.66 ± 1.02	84.96 ± 1.21	83.28 ± 1.36	88.34 ± 0.24	0.412 ± 0.003	0.178 ± 0.001	3.70 ± 0.18
LMF (CE + focal control)	87.73 ± 0.73	81.83 ± 0.72	83.66 ± 1.24	82.37 ± 0.96	87.90 ± 0.75	0.399 ± 0.009	0.176 ± 0.007	3.28 ± 0.56
TMC	80.79 ± 0.82	73.79 ± 3.70	71.22 ± 1.82	70.63 ± 2.22	80.62 ± 0.95	1.422 ± 0.003	0.514 ± 0.001	44.98 ± 0.85
QMF	84.79 ± 1.10	82.68 ± 1.43	77.36 ± 0.86	78.37 ± 0.67	84.45 ± 1.03	0.465 ± 0.013	0.218 ± 0.009	3.69 ± 0.88
Proposed	88.26 ± 0.21	81.37 ± 0.23	84.81 ± 0.30	82.26 ± 0.31	88.24 ± 0.26	0.367 ± 0.013	0.173 ± 0.005	2.76 ± 0.27

The modality matrix identifies the visual stream as the principal clean-input discriminator. Visual-only training reaches 88.98% Accuracy and 83.95% Macro F1; V+S follows at 88.50% and 83.32%, and V+T reaches 88.26% and 82.24%. In contrast, S and T+S attain 19.61/8.28% and 16.06/4.91% Accuracy/Macro F1, respectively. Thus, the fixed text bank structures an observation-conditioned query but does not replace sample-specific appearance, while the archived array is most useful in conjunction with vision. The complete model is designed to control this auxiliary state under clean and degraded conditions; its mechanism contribution is evaluated directly in Sections IV-G, IV-I, and IV-K.

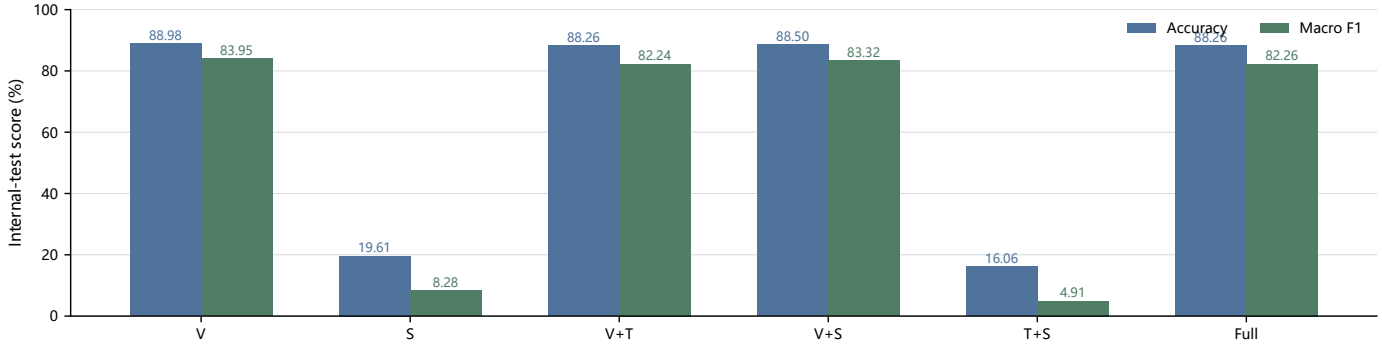


Figure 5. Clean internal-test Accuracy and Macro F1 for the independently retrained modality configurations. Every bar is the mean over training seeds 42, 43, and 44.

Among the six three-modality fusion families, CE-trained LMF is the strongest clean-input Macro-F1 reference. Across three seeds, it reaches 88.30% Accuracy and 83.28% Macro F1, compared with 88.26% and 82.26% for the proposed model. Visual-only remains the strongest clean-input classifier overall at 88.98% Accuracy and 83.95% Macro F1. The objective-matched LMF CE+focal control reaches 87.73% Accuracy and 82.37% Macro F1. Relative to this control, the proposed model is 0.53 pp higher in Accuracy and 0.11 pp lower in Macro F1, with lower NLL (0.367 versus 0.399), Brier score (0.173 versus 0.176), and ECE (2.76% versus 3.28%). Thus, the original LMF clean Macro-F1 advantage is not created by omitting focal loss, and the proposed calibration advantage persists after objective matching. Relative to principal CE-trained LMF, the proposed model also has lower Macro-F1 variation (0.31 versus 1.36 points). In the prespecified seed-42 paired diagnostic, NLL improves by -0.057 with a 95% source-image-cluster interval of [-0.110, -0.010], while the Accuracy and Macro-F1 intervals include zero. The clean-input evidence therefore identifies vision as the strongest clean discriminator and motivates the fault- and visual-difficulty utility tests reported below.

The proposed ensemble reaches 88.98% Accuracy, 83.66% Macro F1, and 1.91% ECE. Relative to the QMF ensemble, the paired differences are +4.60 pp Accuracy with a 95% source-image-cluster interval of [+2.61, +6.57] and +5.25 pp Macro F1 with an interval of [+1.19, +9.80]. The proposed ensemble uniquely corrects 50 objects that QMF misses, whereas QMF uniquely corrects 12; the object-level exact McNemar sensitivity test gives $p = 1.21 \times 10^{-6}$. NLL decreases from 0.460 to 0.363, and ECE decreases from 4.49% to 1.91%. These paired results make the predictive advantage clearer than a comparison of separately averaged seed scores.

The remaining references fill out the comparison spectrum under the same three-seed protocol. Concat-MLP obtains 86.56% Accuracy and 80.49% Macro F1, while GMU reaches 86.84% and 81.93%. The adapted quality-aware QMF baseline [11] reaches $84.79\% \pm 1.10\%$ Accuracy and $78.37\% \pm 0.67\%$ Macro F1; the proposed model improves these means by 3.47 and 3.89 points and lowers NLL by 0.098. The evidential TMC adaptation [21] reaches $80.79\% \pm 0.82\%$ Accuracy and $70.63\% \pm 2.22\%$ Macro F1. These task-adapted models reuse our encoders, inputs, and schedule; they are not presented as verbatim reproductions on the source papers' original datasets.

Confidence-ranked selective evaluation provides a second operational view while retaining all objects in the headline results. Across the full test risk versus coverage curve, the proposed

ensemble has an area under that curve (AURC) of 0.0202 versus 0.0362 for QMF, a 44.2% relative reduction. At 80% test coverage, selective Accuracy is 96.07% for the proposed ensemble and 92.28% for QMF. In a threshold-based analysis, the confidence threshold selected to attain at least 95% Accuracy at maximum validation coverage is 0.576 for the proposed ensemble and 0.590 for QMF. Applied unchanged to the test split, the proposed policy accepts 90.19% of objects at 92.75% Accuracy, while QMF accepts 80.99% at 91.93%. These routing diagnostics complement the all-sample scores in Table IV.

E. Utility under Controlled Visual and Sensor Stress

Table V compares the principal deployment alternatives directly: Visual-only, LMF, and the proposed model. The same three trained checkpoints per family are evaluated without parameter updates. Sensor-state conditions leave RGB unchanged; visual conditions leave the archived sidecar unchanged. Visual-only is invariant to sidecar interventions by construction. The stress transforms are deterministic except for high Gaussian grid noise, for which one fixed perturbation seed is shared across methods. The final two rows are paired low-light counterfactuals on the same proposed checkpoints and identical low-light RGB tensors: one forces availability to zero, and the other applies a fixed global derangement with no sample retaining its own sidecar.

TABLE V. DIRECT THREE-SEED COMPARISON UNDER CONTROLLED INPUT STRESS AND A SAME-CHECKPOINT LOW-LIGHT SIDECAR COUNTERFACTUAL. BOLD DENOTES THE HIGHEST MEAN AMONG THE THREE MODEL FAMILIES WHERE ALL WERE EVALUATED.

Condition	Visual-only Acc./F1 (%)	LMF Acc./F1 (%)	Proposed Acc./F1 (%), NLL
Missing sensor-state, 100%	88.98 ± 0.67 / 83.95 ± 1.06	88.34 ± 0.30 / 83.24 ± 1.47	88.90 ± 0.25 / **84.30 ± 0.03**, 0.362
Gaussian grid noise, $\sigma = 0.15$	88.98 ± 0.67 / 83.95 ± 1.06	88.34 ± 0.28 / 83.27 ± 1.45	88.86 ± 0.32 / 83.36 ± 0.95, 0.363
Directional grid bias, 0.15	88.98 ± 0.67 / 83.95 ± 1.06	88.30 ± 0.25 / 83.27 ± 1.45	88.58 ± 0.35 / 83.34 ± 1.09, 0.363
Stuck grid	88.98 ± 0.67 / 83.95 ± 1.06	88.34 ± 0.30 / 83.24 ± 1.47	88.54 ± 0.14 / 82.72 ± 0.60, 0.373
9×9 average blur	57.67 ± 0.70 / 41.87 ± 0.55	56.50 ± 0.92 / **42.96 ± 1.13**	57.02 ± 0.96 / 42.08 ± 0.57, 1.668
Low light, aligned sidecar	31.23 ± 0.24 / 25.03 ± 1.14	**33.25 ± 2.91** / 26.15 ± 2.07	32.77 ± 0.50 / **27.06 ± 0.12**, 3.024
Low light, sidecar forced missing	-	-	32.36 ± 1.04 / 24.21 ± 0.95, 3.234
Low light, sidecar globally permuted	-	-	32.28 ± 0.39 / 26.21 ± 0.13, 3.087
Central $40\% \times 40\%$ occlusion	**72.84 ± 1.33** / **67.56 ± 1.39**	69.57 ± 0.73 / 65.31 ± 1.14	72.76 ± 1.71 / 65.43 ± 1.84, 0.772

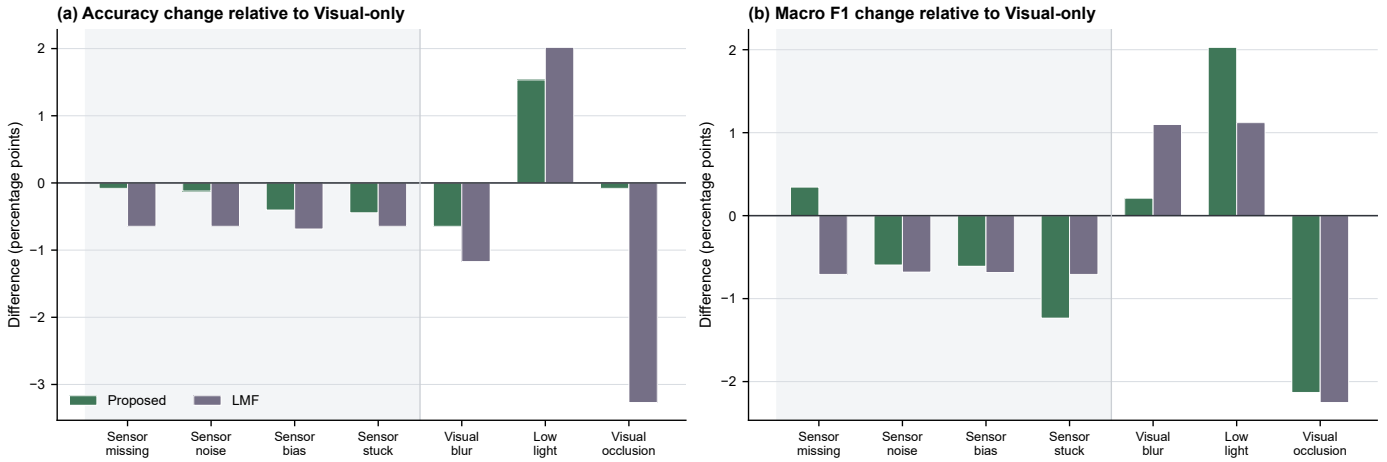


Figure 6. Accuracy and Macro-F1 differences of the proposed model and LMF relative to Visual-only under sensor faults and visual degradations. Values are differences between three-training-seed means; the zero line is the Visual-only reference.

The result gives a bounded answer to why the sidecar path is retained. Visual-only is the strongest clean classifier and remains difficult to exceed under isolated grid faults. Under complete sidecar absence, the proposed model is effectively tied in Accuracy and is 0.35 pp higher in Macro F1, demonstrating a graceful fallback rather than a broad superiority claim. The clearer complementary regime is low light: the proposed model exceeds Visual-only by 1.54 pp Accuracy and 2.03 pp Macro F1 and has lower NLL (3.024 versus 3.170). More importantly, the paired counterfactual holds the proposed checkpoint and low-light RGB fixed. Relative to aligned sidecars, forced absence changes Accuracy/Macro F1/NLL from 32.77/27.06%/3.024 to 32.36/24.21%/3.234, a mean Macro-F1 reduction of 2.85 ± 1.07 pp across training seeds. Global sidecar permutation changes the same metrics to 32.28/26.21%/3.087, reducing Macro F1 by 0.85 ± 0.12 pp. This confirms dependence on sample-sidecar correspondence for the predefined low-light transformation, but the image-conditioned provenance prevents interpreting that dependence as independent physical sensing.

F. Class-Wise Behavior and Error Structure

Figure 7 places class-wise F1 and source-image-cluster intervals beside the long-tailed training distribution. Human-readable object names are used in place of internal dataset keys. The descriptive Pearson correlation between $\log$ training support and seed-42 class F1 is 0.653. Larger classes therefore tend to perform better, but frequency is not the only determinant: the 29-example AA dry-cell battery class reaches 93.10% F1, while the 76-example broken ceramic shard class reaches 83.69% F1. The dashed line marks the seed-42 Macro F1 of 82.44%.

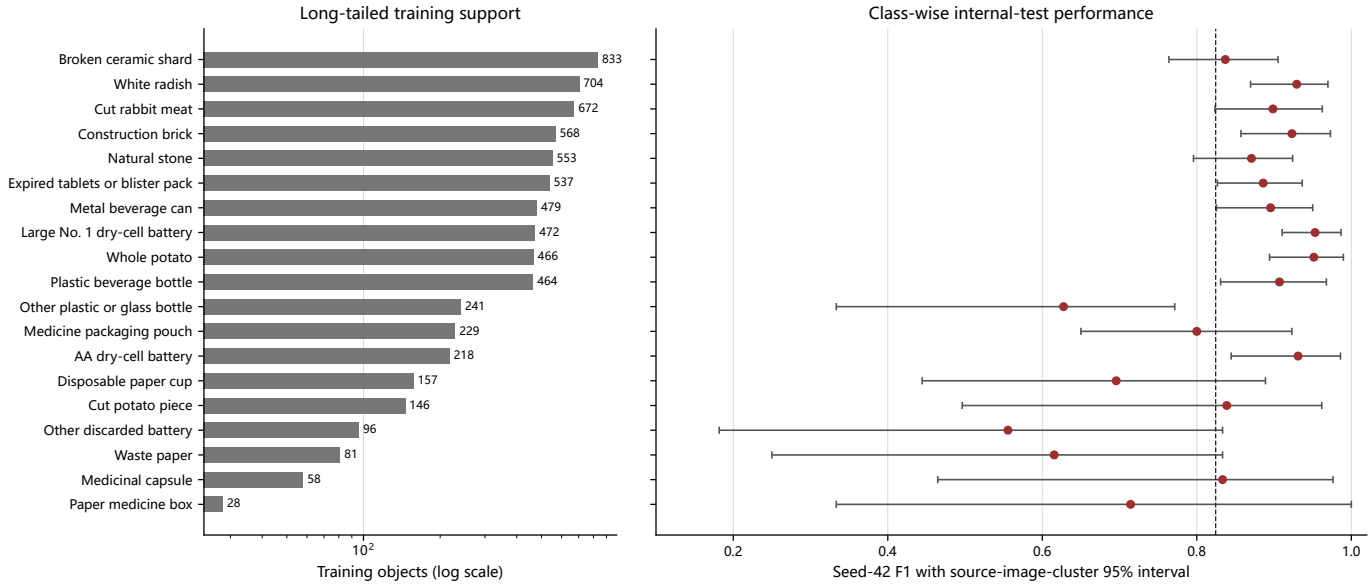


Figure 7. Training support and seed-42 class-wise F1 on the internal test split. Error bars are 95% source-image-cluster percentile-bootstrap intervals; the dashed line is the seed-42 Macro F1.

The highest point estimates occur for large No. 1 dry-cell batteries (95.30% F1), whole potatoes (95.15%), AA dry-cell batteries (93.10%), and white radishes (92.93%). The weakest are other discarded batteries (55.56%), waste paper (61.54%), and other plastic or glass bottles (62.75%). Error structure is consistent with fine-grained category proximity: three of the eleven other discarded batteries are assigned to the large No. 1 dry-cell battery category, while errors for other plastic or glass bottles are distributed across several visually related or low-support categories. The model also predicts all five paper medicine boxes correctly, but the category's 71.43% F1 and wide interval reflect four false-positive predictions and the small support.

These intervals add essential context to rare-class point estimates. For example, F1 intervals span approximately 18.18-83.33% for other discarded batteries, 25.00-83.33% for waste paper, and 33.33-100.00% for paper medicine boxes. The class plot characterizes the long-tail structure and explains why Accuracy and Weighted F1 remain near 88% while Macro F1 is lower: frequent classes dominate object-level correctness, whereas every rare class contributes equally to the macro average.

G. Matched Reliability/Fallback-Subsystem Ablation

The primary ablation jointly removes sensor-latent prediction, reliability gating, raw-grid and predicted-grid quality evidence, and their associated auxiliary losses. It retains the three branch encoders, attention fusion, training degradation, optimizer, data, and 100-epoch schedule. Each ablation run is retrained with the seed matched to the full model. This comparison measures the contribution of the compound subsystem, while Section IV-H refines the analysis at the loss-component level.

As shown in Table VI, the full model improved Accuracy by 0.81 ± 0.49 percentage points (pp) and Macro F1 by 0.51 ± 0.23 pp across seeds. Both differences were positive for all three seeds. The equal-protocol ensemble comparison increases these paired differences to 1.57 pp Accuracy and 1.46 pp Macro F1. The Accuracy interval [0.47, 2.82] excludes zero and the exact McNemar sensitivity test gives $p = 0.0146$; the Macro F1 interval [-0.53, 3.72] is wider. Checkpoint aggregation therefore strengthens the internal classification evidence, with seed-level variability retained as a complementary view.

TABLE VI. MATCHED COMPOUND RELIABILITY/FALLBACK-SUBSYSTEM ABLATION. INTERVALS ARE PAIRED SOURCE-IMAGE-CLUSTER BOOTSTRAP INTERVALS.

Seed	Full minus ablation Accuracy, pp (95% CI)	Full minus ablation Macro F1, pp (95% CI)	Exact McNemar p
42	+0.36 [-0.94, +1.68]	+0.53 [-1.51, +3.00]	0.7201
43	+0.73 [-0.58, +2.13]	+0.27 [-2.49, +3.32]	0.3616
44	+1.33 [+0.12, +2.62]	+0.73 [-1.01, +2.61]	0.0614
Mean ± sample SD	+0.81 ± 0.49	+0.51 ± 0.23	-
Validation-selected ensemble	+1.57 [+0.47, +2.82]	+1.46 [-0.53, +3.72]	0.0146

The no-reliability ensemble reaches 87.41% Accuracy and 82.21% Macro F1, compared with 88.98% and 83.66% for the complete ensemble. The complete system uniquely corrects 19 objects while the ablation uniquely corrects six. A complementary inference intervention bypasses the compound fallback/reliability path in the fixed seed-42 full checkpoint. Keeping that complete path improves Accuracy by 2.78 pp and Macro F1 by 3.50 pp, with paired cluster intervals [1.23, 4.33] and [1.46, 5.89] pp. The matched retraining and path intervention together show both compensatory training behavior and active use of the learned subsystem; they do not isolate the dynamic coefficient from its fallback endpoint.

H. Layered Ablation Analysis

Table VII separates gate adaptivity, two auxiliary objectives, and one gradient-routing choice while retaining the rest of the full architecture. The constant-gate control keeps the complete fallback and fusion paths but replaces the sample-wise coefficient by $r_{const} = a\ \mathrm{sigmoid}\ (b_r)$. Removing clean-grid reconstruction tests whether the explicit spatial prediction target contributes beyond latent reconstruction. Removing pairwise ranking tests whether ordering examples by injected quality adds information beyond weighted pointwise calibration. The no-stop-gradient control allows the reliability objective to update the predicted-grid decoder through the comparison features. All variants were

independently retrained for seeds 42, 43, and 44 under the same 100-epoch budget and validation-Accuracy checkpoint rule.

TABLE VII. THREE-SEED GATE-ADAPTIVITY, LOSS-COMPONENT, AND GRADIENT-ROUTING ABLATIONS ON THE FIXED INTERNAL BENCHMARK.

Configuration	Accuracy, mean ± SD (%)	Macro F1, mean ± SD (%)	NLL	Brier	ECE (%)
Full model	88.26 ± 0.21	82.26 ± 0.31	0.367	0.173	2.76
Constant learned gate	88.46 ± 0.43	82.44 ± 0.52	0.368	0.172	2.64
No reliability/fallback subsystem	87.45 ± 0.56	81.75 ± 0.52	0.383	0.181	2.86
No clean-grid reconstruction	87.73 ± 0.30	81.63 ± 0.60	0.379	0.178	2.92
No reliability-ranking loss	87.73 ± 0.60	81.65 ± 0.76	0.369	0.174	2.91
No predicted-grid stop-gradient	88.10 ± 0.30	82.46 ± 0.29	0.373	0.174	3.11

Removing grid reconstruction reduced mean Accuracy by 0.53 pp and Macro F1 by 0.63 pp, while worsening NLL and Brier score. Removing ranking produced a similar mean classification reduction, while its NLL and Brier score stayed close to the full model and its training variability increased. The results suggest that the grid objective mainly strengthens the fallback representation, whereas ranking mainly organizes the gate across fault severities. Their interaction with calibration and latent reconstruction makes the reported differences joint rather than additive effects.

Allowing reliability gradients to update the predicted-grid decoder changes mean Accuracy by -0.16 pp and Macro F1 by +0.20 pp relative to the detached design. Its NLL, Brier score, and ECE are 0.373, 0.174, and 3.11%, compared with 0.367, 0.173, and 2.76% with stop-gradient. The routing choice therefore improves mean Accuracy and calibration while preserving a cleaner separation between prediction and comparison; it is not claimed to improve every classification metric.

Matched constant-gate retraining does not establish a consistent average classification advantage for sample-adaptive r. On clean input, dynamic minus constant is -0.20 ± 0.28 pp Accuracy and -0.17 ± 0.81 pp Macro F1; dynamic NLL is slightly lower (0.367 versus 0.368), whereas constant Brier/ECE are 0.172/2.64% versus 0.173/2.76%. The dynamic gate is selectively beneficial under strong Gaussian noise, giving $+0.48 \pm 0.45$ pp Accuracy, $+0.98 \pm 0.91$ pp Macro F1, and a 0.0055 NLL reduction. Its Macro-F1 differences are $+0.35 \pm 1.31$ pp for complete absence, $+0.11 \pm 1.78$ pp for directional bias, $+0.23 \pm 0.54$ pp for a stuck grid, and -0.21 ± 0.48 pp for mixed faults. ECE improves for dynamic gating in every fault row by 0.13–0.66 pp, but not on clean input. The matched evidence therefore assigns the primary classification-robustness result to the fallback subsystem, while r supplies fault-sensitive quality ordering, fault-conditioned calibration, and a selective high-noise benefit.

The remaining component results clarify how the compound subsystem is trained. The full dynamic model has the best mean Accuracy among the loss and routing controls and the lowest NLL and Brier score among those controls, while the two objectives provide complementary spatial-prediction and reliability-ordering supervision. The constant-gate row is interpreted separately because it isolates sample adaptivity rather than deleting either objective.

A second ablation layer measures active path reliance without allowing the remaining modules to compensate through retraining. Each intervention is applied to all three independently trained full checkpoints, and Table VIII reports the mean and sample SD across seeds. This protocol complements the retrained comparisons: it asks which learned paths the selected full models actually use at inference.

TABLE VIII. THREE-CHECKPOINT INFERENCE-PATH INTERVENTIONS. ALL ROWS USE THE SAME SEED-42, SEED-43, AND SEED-44 FULL CHECKPOINTS WITHOUT RETRAINING.

Active inference configuration	Accuracy, mean ± SD (%)	Macro F1, mean ± SD (%)	Full minus intervention Accuracy (pp)	Full minus intervention Macro F1 (pp)
Full model	88.26 ± 0.21	82.26 ± 0.31	0.00	0.00
Visual path only	86.40 ± 1.33	80.35 ± 2.43	+1.86	+1.92
Visual + text paths	87.89 ± 0.24	82.06 ± 0.47	+0.36	+0.20
Visual + sensor paths	86.08 ± 0.21	80.53 ± 0.44	+2.18	+1.74
Reliability/fallback path bypassed	86.44 ± 1.03	80.20 ± 1.16	+1.82	+2.07
Reliability evidence replaced by learned bias	88.70 ± 0.25	82.90 ± 0.54	-0.44	-0.63
FiLM path bypassed	84.91 ± 1.21	78.09 ± 1.99	+3.35	+4.18
Attention replaced by equal-mean fusion	85.39 ± 0.25	80.17 ± 0.11	+2.87	+2.09

The visual-only result quantifies the net contribution of the nonvisual paths, while the two bimodal rows separate class-text and archived-sidecar access. The fixed-checkpoint interventions show path reliance rather than independently retrained component contributions: the trained checkpoints exhibit 4.18-pp Macro-F1 reliance on FiLM and 2.09-pp reliance on attention under the stated bypasses. Bypassing the complete reliability/fallback path reduces mean Accuracy and Macro F1 by 1.82 and 2.07 pp, respectively, which is larger than the 0.81/0.51 pp retrained difference because the fixed checkpoint cannot relearn a compensating fusion rule.

Replacing all learned reliability evidence by the gate head's learned bias gives 88.70% Accuracy and 82.90% Macro F1 on clean inputs, 0.44 and 0.63 pp above the intact dynamic gate. Under mixed faults, the same fixed-bias intervention reaches 88.55% Accuracy and 82.82% Macro F1 versus 88.37% and 82.61% for the intact gate after hierarchical averaging over three training seeds and five fault realizations. The dynamic score is therefore supported as a fault-sensitive quality ordering variable by Section IV-K, but this intervention does not attribute an independent average classification gain to evidence-dependent

variation in r. The positive whole-path results belong to the compound fallback-and-control subsystem.

The reliability evidence was also masked at finer granularity. Across individual and grouped removals of raw-grid statistics, predicted-observed grid comparison, latent discrepancy, latent consistency, variance, and learned-head availability, the maximum absolute clean-input change was 0.44 pp Accuracy and 0.63 pp Macro F1; under mixed faults it was 0.19 and 0.21 pp after averaging five matched perturbation seeds within each training seed. This bounded sensitivity demonstrates overlapping evidence: individual cues can substitute for one another, whereas bypassing the complete reliability/fallback path produces a consistent two-point Macro-F1 effect.

I. Robustness to Controlled Sensor Degradation

Table IX compares the full and matched ablation models under clean input and five predefined sensor-fault conditions. Each seed-42/43/44 trained model is first averaged over five matched perturbation seeds (2026-2030), after which the mean and sample SD are computed across the three training seeds. The full model has positive mean Accuracy, Macro F1, and ECE effects in all six conditions. The largest Macro-F1 effect occurs under complete sensor absence ($+2.37 \pm 0.66$ pp), while strong Gaussian noise gives $+1.59 \pm 0.76$ pp. This pattern aligns with the intended prediction fallback and residual suppression.

TABLE IX. THREE-TRAINING-SEED FULL-MODEL IMPROVEMENT OVER THE MATCHED ABLATION UNDER CONTROLLED SIDECAR DEGRADATION. FAULT ROWS FIRST AVERAGE FIVE MATCHED PERTURBATION SEEDS; THE DETERMINISTIC CLEAN ROW REUSES THE BATCH-16, ZERO-WORKER PREDICTIONS REPORTED IN TABLE VI.

Condition	Full minus ablation Accuracy, mean ± SD (pp)	Full minus ablation Macro F1, mean ± SD (pp)	ECE improvement, mean ± SD (pp)
Clean	+0.81 ± 0.49	+0.51 ± 0.23	+0.10 ± 0.25
Missing 100%	+1.49 ± 0.49	+2.37 ± 0.66	+0.44 ± 0.32
Gaussian noise, $\sigma = 0.15$	+1.46 ± 0.51	+1.59 ± 0.76	+0.28 ± 0.29
Directional bias, amplitude 0.08	+0.86 ± 0.27	+0.84 ± 0.54	+0.68 ± 0.64
Stuck grid	+1.25 ± 0.50	+1.02 ± 1.06	+0.37 ± 0.10
Mixed faults	+0.97 ± 0.44	+1.06 ± 0.78	+0.29 ± 0.14

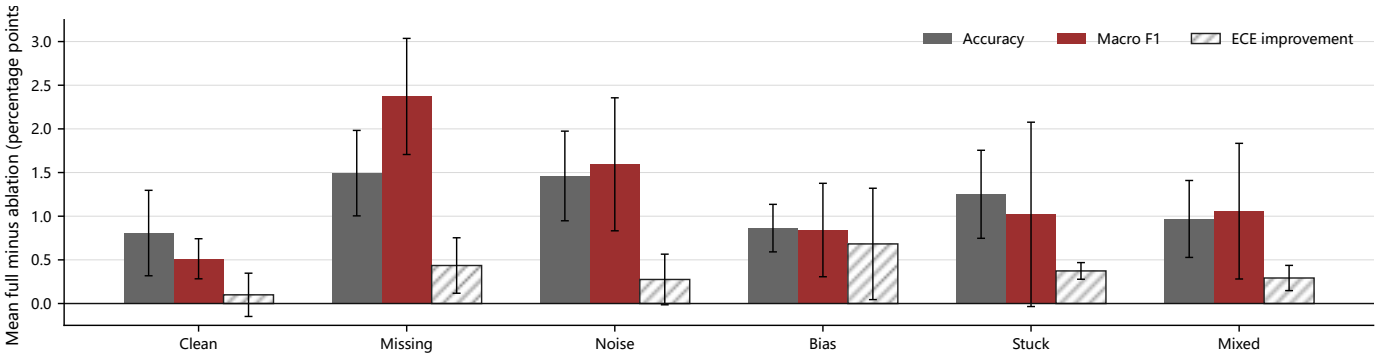


Figure 8. Mean difference between the full model and matched reliability/fallback ablation across three training seeds under controlled sensor faults. Error bars show the sample SD across training seeds. Positive ECE improvement means lower ECE for the full model.

Figure 8 makes two patterns visible. First, the classification advantage is larger under missing and noisy inputs than under the clean condition, which is the expected regime for a fallback-and-attenuation mechanism. Second, mean ECE improvement is positive in every condition, although its cross-training-seed variation is wider for bias. The result extends the fault comparison from one selected checkpoint to three independently trained model pairs without changing the direction of the classification effects.

The absolute full-model scores are not monotone in apparent fault severity: some perturbed conditions slightly exceed the clean result. This can occur when the archived sidecar branch is imperfect for a subset of objects and a perturbation causes the gate to rely more heavily on the learned fallback. We therefore interpret the intervention as a comparative mechanism test, not as evidence that corruption improves the auxiliary state or that stronger perturbations are harmless. Complete absence is the cleanest causal test of the fallback equation because $a = 0$ deterministically removes the archived residual.

Together, these interventions provide an internal mechanism profile for the archived sidecar interface. A future hardware study would require newly collected paired measurements, device and calibration metadata, and a locked cross-device protocol; those conditions are not represented by the current archive.

J. Developmental Class-Memory Fallback Probe

Table X tests what the developmental memory lookup actually contributes under complete sensor absence. Correct descriptions reduce latent MSE and grid MAE by 92.0% and 10.3% relative to vision-conditioned fallback, and by 28.1% and 14.0% relative to shuffled descriptions. The scale-aware results lead to the same interpretation: latent R^2 changes from -9.290 for vision-conditioned prediction to 0.192 for correct descriptions, while grid R^2 rises from 0.157 to 0.251. These gains establish effective class-memory binding, but not a language-specific advantage.

TABLE X. DEVELOPMENTAL CLASS-MEMORY CONTROLS ON THE REUSED INTERNAL-TEST SPLIT UNDER COMPLETE SENSOR ABSENCE. THE FIRST FOUR ROWS ARE PAIRED INFERENCE INTERVENTIONS; THE CLASS-ID ROW USES SEPARATELY TRAINED, PROTOCOL-MATCHED CHECKPOINTS. VALUES ARE MEAN ± SAMPLE SD OVER TRAINING SEEDS; LOWER MSE/NMSE/MAE AND HIGHER R^2 ARE BETTER.

Fallback condition	Missing Acc. (%)	Missing Macro F1 (%)	Latent MSE	Latent NMSE / R^2	Grid MAE	Grid R^2
Vision-conditioned	89.55 ± 0.18	84.83 ± 0.69	4.031e-4 ± 7.5e-5	10.290 / -9.290	0.2496 ± 0.0052	0.157
Correct descriptions	89.43 ± 0.14	84.70 ± 0.92	3.212e-5 ± 3.1e-6	0.808 / 0.192	0.2239 ± 0.0005	0.251
Class names only	89.47 ± 0.21	84.77 ± 1.01	3.282e-5 ± 3.1e-6	0.825 / 0.175	0.2275 ± 0.0009	0.243
Shuffled descriptions	89.51 ± 0.28	84.79 ± 1.05	4.468e-5 ± 4.2e-6	1.124 / -0.124	0.2604 ± 0.0079	0.061
Fixed orthogonal class-ID keys	89.47 ± 0.12	84.78 ± 0.88	**3.165e-5 ± 3.1e-6**	**0.796 / 0.204**	**0.2228 ± 0.0006**	**0.254**

Classification changes are below 0.13 pp and do not favor correct descriptions. More decisively, the nonsemantic class-ID keys slightly outperform description keys in latent MSE, latent R^2, grid MAE, and grid R^2. Shuffling a trained description-to-memory association is harmful, but preserving the class index with orthogonal keys is sufficient to recover the gain. The supported conclusion is therefore **class-aligned memory retrieval**, not an irreplaceable contribution from natural-language semantics. Because the internal benchmark informed this refinement and the class-ID control required separate short training, Table X is developmental mechanism evidence rather than an external confirmatory result.

K. Direct Reliability-Score Behavior

Figure 9 and Table XI evaluate the final reliability coefficient r itself using all three independently trained full-model checkpoints. Mean r decreases from 0.857 ± 0.005 for clean input to 0.838 ± 0.005, 0.801 ± 0.003, and 0.760 ± 0.002 for low, medium, and high noise. The corresponding observation-level severity correlation is $\rho = -0.521 \pm 0.033$ across training seeds. Directional bias produces a second ordered response, from 0.857 ± 0.005 to 0.839 ± 0.005 and 0.822 ± 0.005, with $\rho = -0.240 \pm 0.013$. Complete absence is an exact boundary case: availability forces every score to zero.

TABLE XI. DIRECT BEHAVIOR OVER THREE INDEPENDENTLY TRAINED RELIABILITY GATES. METRICS ARE FIRST AVERAGED OVER FIVE FAULT REALIZATIONS WITHIN EACH CHECKPOINT AND THEN SUMMARIZED AS MEAN ± SAMPLE SD OVER TRAINING SEEDS. MEDIANS ARE AVERAGED OVER CHECKPOINTS. AUROC/AUPRC USE EFFECTIVE CHANGED RECORDS, WITH AUPRC BALANCED TO EQUAL CLEAN/FAULT TOTAL WEIGHT.

Sensor condition	Mean r	Median r	Effective faults (%)	AUROC	Balanced AUPRC
Clean	0.857 ± 0.005	0.864	–	–	–
Missing 100%	0.000	0.000	100.0	1.000 ± 0.000	1.000 ± 0.000
Noise, low (0.03)	0.838 ± 0.005	0.846	100.0	0.591 ± 0.005	0.611 ± 0.006
Noise, medium (0.08)	0.801 ± 0.003	0.803	100.0	0.751 ± 0.017	0.781 ± 0.012
Noise, high (0.15)	0.760 ± 0.002	0.759	100.0	0.864 ± 0.019	0.889 ± 0.017
Bias, low (0.08)	0.839 ± 0.005	0.846	100.0	0.589 ± 0.004	0.599 ± 0.009
Bias, high (0.15)	0.822 ± 0.005	0.825	100.0	0.668 ± 0.009	0.682 ± 0.003
Stuck, first-cell latch	0.835 ± 0.009	0.854	62.6	0.677 ± 0.031	0.637 ± 0.025
Mixed faults	0.774 ± 0.003	0.814	71.2	0.789 ± 0.012	0.772 ± 0.007

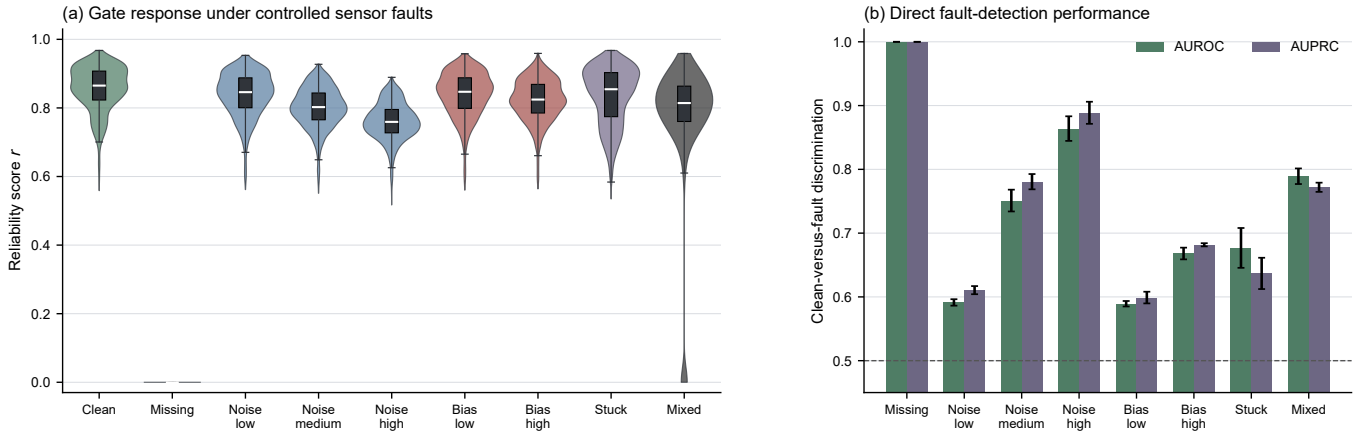


Figure 9. Distribution of the learned reliability score and its direct clean-versus-effective-fault discrimination over three independently trained models. Missing input is forced to zero by availability; progressively stronger noise and bias produce ordered score reductions.

Across all eight fault conditions, macro AUROC is 0.741 ± 0.011 and balanced macro AUPRC is 0.746 ± 0.006; excluding deterministic missingness gives 0.704 ± 0.013 and 0.710 ± 0.007. High noise and mixed faults are distinguished most clearly among available inputs. The first-cell latch changes only 62.6% of test records because 309 of 826 clean arrays are already spatially uniform and therefore remain identical after replication. Evaluating only records that change keeps the gate diagnostic tied to observable sensor change rather than the scenario name alone.

The direct score profile complements the classification robustness result in Section IV-I. The latter asks whether gating improves decisions; the present experiment asks whether r identifies degraded observations. Its monotone severity response, exact missing boundary, and above-reference fault discrimination establish that the gate performs sample-level quality ordering, while the family-specific values preserve the resolution of the controlled protocol.

L. Variance Diagnosis

Post-hoc variance analysis used 826 objects and the 512-dimensional sensor latent space. For the full model, the Spearman correlation between per-sample mean predicted variance and latent reconstruction MSE was 0.892. Relative to the preceding variance-prior model, the increase was 0.070 with a paired source-image-cluster interval of [0.0477, 0.0954]. Thus, predicted variance provides useful relative error ranking. Mean predicted variance is 44.3 times the observed latent MSE, and nearly 100% of standardized residuals fall within ± 1.96, indicating an over-dispersed absolute scale. We accordingly use the variance head as a ranking diagnostic for sensor-reliability gating.

The distinction between ranking and calibration is operationally important. The high Spearman coefficient shows that difficult samples usually receive larger predicted variance, helping the gate order observations. The 44.3-fold scale ratio and near-100% nominal 95% coverage motivate future calibration through a revised variance range or prior followed by retraining and validation.

The present objective consequently uses variance in two bounded ways: λ is restricted to $[-3, 2]$, and its aggregate enters the reliability head alongside directly observed grid and consistency features. This preserves reconstruction-error ranking while limiting the influence of any single uncertainty scalar.

M. Computational Profile and Deployment Boundary

The full model contains 77,467,164 parameters, of which 57,896,764 are trainable; the reliability/fallback subsystem adds 100,322 total and 1,675,427 trainable parameters over the matched ablation. At batch 16, the seed-42 evaluation measured 1.754 ms per sample (570 samples/s) after three warm-up batches. This CUDA model-forward measurement excludes data loading, preprocessing, transfer, sidecar construction, and batch-1 deployment latency, so it is reported as an environment-specific throughput estimate rather than an end-to-end deployment benchmark.

The small total-parameter increment shows that the pretrained visual and text backbones account for most storage. Its trainable-parameter difference is larger because the ablation changes which connected modules participate in optimization. Runtime is governed mainly by image feature extraction and text encoding rather than the scalar quality heads. In a deployed closed-set system, fixed class prototypes also permit straightforward caching.

The principal model and archived baseline runs used batch size 16 and zero data-loader workers. The matched constant-gate control retains batch size 16 but uses the audited four-worker prefetch path described in Section IV-C; its sampler preserves the zero-worker sample order. This operational change reduces host-side stalls without changing the reported evaluation runtime, which remains batch 16 with zero workers for every dynamic-versus-constant comparison.

End-to-end latency additionally includes image acquisition, object cropping, preprocessing, sidecar construction, decision transfer, and actuator response. The reported 570 samples/s therefore characterizes the neural forward pass on the stated desktop GPU, providing a reproducible compute reference for later system integration.

V. DISCUSSION

A. What the Evidence Supports

The experiments establish a bounded operating regime for reliability-gated sidecar fusion. Vision is the principal clean-input discriminator: Visual-only obtains 88.98% Accuracy and 83.95% Macro F1. Among the three-modality families, CE-trained LMF records the highest clean Macro F1, whereas the proposed model is within 0.04 pp of LMF in Accuracy and has lower NLL, Brier score, and ECE. The objective-matched LMF CE+focal control reaches 87.73/82.37% Accuracy/Macro F1; Proposed reaches 88.26/82.26% and retains lower NLL, Brier, and ECE. The sidecar path becomes complementary under the predefined low-light stress, where the proposed model exceeds Visual-only by 1.54 pp Accuracy and 2.03 pp Macro F1. Under complete sidecar absence, the two are effectively tied in Accuracy and the proposed model is 0.35 pp higher in Macro F1. The practical claim is therefore conditional complementarity with a defined fallback path, rather than uniform superiority over RGB alone.

The mechanism evidence is consistent across complementary protocols. Independent retraining attributes mean gains of 0.81 pp Accuracy and 0.51 pp Macro F1 to the compound reliability/fallback subsystem. Across three trained model pairs, the full-minus-ablation Macro-F1 effect increases from 0.51 ± 0.23 pp on the deterministic clean evaluation to 2.37 ± 0.66 pp under complete sidecar absence after averaging five matched fault realizations within each model. Fixed-checkpoint bypasses show active reliance on the fallback/reliability path, FiLM, and attention, with Macro-F1 differences of 2.07, 4.18, and 2.09 pp, respectively. These numbers measure path reliance under intervention, not retrained component contributions.

The matched constant-gate control narrows the causal attribution further. It slightly exceeds the dynamic model on clean Accuracy/Macro F1 by 0.20/0.17 pp and under mixed faults by 0.19/0.21 pp, whereas the dynamic gate leads by 0.48/0.98 pp under strong Gaussian noise. Dynamic gating also lowers ECE in every evaluated fault condition by 0.13–0.66 pp, despite a 0.13-pp clean ECE disadvantage. These mixed classification directions do not support a general average gain from sample adaptivity. Instead, the fallback subsystem accounts for the principal classification robustness, while the learned score provides fault-sensitive ordering and selectively improves high-noise prediction and fault-conditioned calibration.

The predicted-grid stop-gradient control sharpens the optimization interpretation. Removing stop-gradient changes Accuracy/Macro F1 by -0.16/+0.20 pp and worsens NLL, Brier score, and ECE, so the detached route is supported for responsibility separation and calibration rather than as a universal classification gain. In a fixed-checkpoint intervention, replacing sample-wise evidence by the gate head's learned output bias slightly exceeds the intact dynamic path in average clean and mixed-fault classification. This is a path-reliance observation, not a matched training comparison. The learned coefficient is independently supported as a fault-sensitive quality-ordering signal by its available-fault AUROC/AUPRC of $0.704 \pm 0.013/0.710 \pm 0.007$ over three trained models.

B. Sensor-State Records and the Class-Memory Probe

The two nonvisual sources have distinct roles. In the proposed 100-epoch model, the 4×4 sidecar is a sample-indexed normalized state whose 16 ordered cells form the archived spatial interface; fixed CLIP prototypes provide a soft class context t_v to the learned fallback predictor and fusion path. Weight, temperature, moisture, conductivity, and odor response remain archived attributes and are not classifier inputs. The residual formulation treats record presence and estimated consistency separately, allowing the same path to retain an aligned state,

attenuate a degraded record, or use μ_{base} when the record is absent. Because current metadata identifies the sidecars as image-conditioned estimates, this mechanism does not establish independence from RGB.

The class-memory study is a separate developmental probe. Correct description keys reduce latent MSE by 92.0% relative to vision-conditioned retrieval and improve latent R^2 from -9.290 to 0.192. Shuffling class-memory bindings degrades reconstruction, but fixed orthogonal class-ID keys slightly exceed description keys on latent MSE, latent R^2, grid MAE, and grid R^2. The supported result is therefore class-aligned memory retrieval, not an irreplaceable contribution from natural-language content. Keeping this probe separate from the principal checkpoints makes the main model definition and evidence boundary explicit.

C. Meaning of the Reliability Score

The learned score r measures agreement with the archived unperturbed state under the defined corruption process. It is neither a device-health probability nor a direct estimate of classification benefit: an unaltered grid can be weakly informative for a category, and a mildly altered grid can still support the correct decision. The constant-gate result makes this distinction empirical rather than merely definitional: better fault ordering and calibration need not translate into a higher average classification score in every condition. Predicted variance serves a related but narrower role; its 0.892 Spearman correlation with latent reconstruction error supports ranking, while the 44.3-fold scale ratio prevents interpreting it as an absolutely calibrated physical variance.

The gate distributions exhibit the intended ordering. Relative to the three-checkpoint clean mean, low, medium, and high noise reduce r by 0.018, 0.056, and 0.096; low and high bias reduce it by 0.018 and 0.035. Noise- and bias-severity correlations are -0.521 ± 0.033 and -0.240 ± 0.013. Complete absence is reported separately because availability sets $r = 0$ exactly. For the first-cell latch, 37.4% of clean grids are already spatially uniform and therefore unchanged by the intervention; separating effective from nominal changes keeps the diagnostic tied to observable sensor evidence.

D. Long-Tail Recognition and Decision Policy

Training support differs by almost 30-fold, and the smallest test classes contain five to eleven objects. Source-image-cluster intervals therefore accompany class-wise scores, while the gap between Accuracy and Macro F1 is interpreted through both scarcity and label proximity. More specialized long-tail objectives, including class-balanced loss, decoupled training, logit adjustment, and calibrated-prior methods [33] [34] [35] [36], define a controlled extension of the present protocol rather than evidence claimed by the current experiments.

Selective prediction provides an immediate deployment policy without changing the classifier. The protocol-matched proposed ensemble has a test AURC of 0.0202, 44.2% below the QMF ensemble, and its confidence-ranked top 80% reaches 96.07% Accuracy. A threshold selected only on validation predictions for a 95% target accepts 90.19% of test objects at 92.75% Accuracy. Thus, uncertain objects can be routed to a secondary sensing or decision stage while the principal model remains the single-checkpoint reference.

E. Scope and Transfer Protocol

The evidence covers grouped RGB sources, archived sidecar records, three training seeds, controlled visual and sidecar interventions, class-binding controls, and direct gate diagnostics. It supports the current 19-class internal benchmark and normalized 4×4 record interface. Transfer to an independently measured device remains a separate study requiring synchronized acquisition, device identifiers, units, calibration references, and a locked external evaluation set.

Reproducibility is maintained by associating each reported run with its manifest, configuration, checkpoint, and prediction file. The three training histories remain independent, and the post-hoc ensemble is labeled as a three-forward-pass deployment variant rather than a capacity-matched single model.

VI. CONCLUSION

This work presented reliability-gated visual-sensor-state fusion with a learned fallback for waste classification. The proposed 100-epoch model combines YOLO11 multiscale visual features, a soft context over CLIP class prototypes, and an archived normalized 4×4 sidecar. Visual and class-text context predict μ_{base} and a reference grid; availability, grid statistics, predicted-record discrepancy, latent agreement, and bounded variance regulate the archived residual. This single path retains a consistent record, attenuates a degraded one, and reaches the learned fallback exactly when the sidecar is absent.

On the 8,654-object internal benchmark, Visual-only remains the strongest clean classifier at 88.98% Accuracy and 83.95% Macro F1. The proposed model obtains $88.26\% \pm 0.21\%$ Accuracy and $82.26\% \pm 0.31\%$ Macro F1, improving on adapted QMF by 3.47 and 3.89 pp. CE-trained LMF has the highest clean Macro F1 among three-modality fusion families; its objective-matched CE+focal control reaches 87.73/82.37% Accuracy/Macro F1, compared with 88.26/82.26% for Proposed, whose NLL, Brier score, and ECE remain lower. Under controlled low light, the proposed model exceeds Visual-only by 1.54 pp Accuracy and 2.03 pp Macro F1. On the same low-light tensors and checkpoints, aligned, forced-missing, and globally permuted sidecars give 27.06%, 24.21%, and 26.21% Macro F1, respectively. This confirms use of sample-sidecar correspondence without establishing an independent physical modality. The compound reliability/fallback subsystem gives +0.81 pp Accuracy and +0.51 pp Macro F1 under matched three-seed retraining. A matched constant-gate control is 0.20/0.17 pp higher on clean Accuracy/Macro F1 and 0.19/0.21 pp higher under mixed faults, while dynamic gating is 0.48/0.98 pp higher under strong Gaussian noise and improves ECE in all fault conditions. Thus, matched retraining assigns the principal classification robustness to fallback and supports r as fault-sensitive quality ordering with selective high-noise utility, not as a uniformly superior classifier. Across three full/ablation model pairs and five fault seeds, the compound subsystem's Macro-F1 effect reaches 2.37 ± 0.66 pp under complete sidecar absence. For available corruptions, direct gate diagnostics over three independently trained models give macro AUROC/AUPRC of $0.704 \pm 0.013/0.710 \pm 0.007$; missingness is handled separately through deterministic availability. The developmental class-memory probe supports class-aligned retrieval, while its nonsemantic class-ID control rules out a language-specific claim.

Together, the modality matrix, visual and sidecar stress tests, matched ablations, fixed-checkpoint path interventions, gate distributions, and calibration analysis characterize where the design adds value and how it behaves. Within the evaluated scope, the method provides a reproducible way to regulate a structured auxiliary state without making its availability a prerequisite for inference.

A. Scope and Data Provenance

The benchmark contains 8,654 annotated records grouped from 5,990 RGB source images. Each object is paired by `sample_id` with one normalized 4×4 sidecar. Current provenance metadata identifies 8,639 records as image-conditioned vision-language estimates and 15 as deterministic fallbacks after structured-output failure. Hashed source-image grouping prevents source overlap across partitions. Weight, temperature, moisture, conductivity, and

odor response are archived as sidecar fields and are not classifier inputs.

The archived `normalized_grid` is finite, strictly 4×4, and bounded to $[0, 1]$. The available records do not contain device identifiers, physical units, site spacing, sampling settings, or raw zero/full-scale references; no unrecorded hardware calibration is inferred. The latent branch standardizes each array to emphasize spatial shape, while the unstandardized bounded values remain available to the reliability heads. The classifier receives a 480×480 RGB crop and the normalized single-channel array; it receives neither camera depth nor the five scalar attributes. Controlled missing, noise, bias/drift, stuck-grid, and mixed conditions are reproducible software interventions on these archived arrays. The principal comparisons use independent seed-42/43/44 checkpoints, whereas the class-memory study and the three-checkpoint ensemble are explicitly reported as separate analyses.

B. Generative AI Use Disclosure

An image-conditioned vision-language workflow generated 8,639 sensor-state sidecars from object crops before model training; a deterministic rule generated the remaining 15 records after structured-output validation failed. OpenAI Codex (GPT-5 series, service version accessed August 11, 2026) was used for language, structure, formatting, experiment scripting, and vector-figure scripting assistance. The reported metrics were produced by the archived checkpoints and evaluation code rather than supplied by the language model.

Human authors checked the manuscript text, citations, numerical values, tables, figures, and claims and retain responsibility for them. The workflow did not modify the RGB images, category labels, trained checkpoints, per-sample model predictions, or metric calculations. AI systems are not authors.

C. Anonymous Code and Data Availability

For double-blind review, an anonymized package will be provided through a venue-approved channel. Subject to redistribution rights, it will contain code, configurations, split manifests, the sensor schema, integrity hashes, predictions, and evaluation scripts; where raw inputs cannot be redistributed, provenance records and hashes will preserve traceability. Identity-bearing acknowledgments and an archival URL will be added after review.

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